Prefix & Notation

Metric Prefix	Prefix	Engineering Notation
Tera	T	10^{12}
Giga	G	10^{9}
Mega	M	10^{6}
Kilo	K	10^{3}
Milli	m	10^{-3}
Micro	µ	10^{-6}
Nano	n	10^{-9}
Pico	P	10^{-12}

Scientific notation

$a \times 10^{b}$

where $1 \le a < 10$, b is non-zero integer, can be either positive or negative

Engineering notation

$a \times 10^{b}$

where $0 \le a$, b can be either positive or negative but must be multiple of 3.

Engineering Prefix (notation)

substitute a prefix in place of the x10 and exponent in engineering notation.

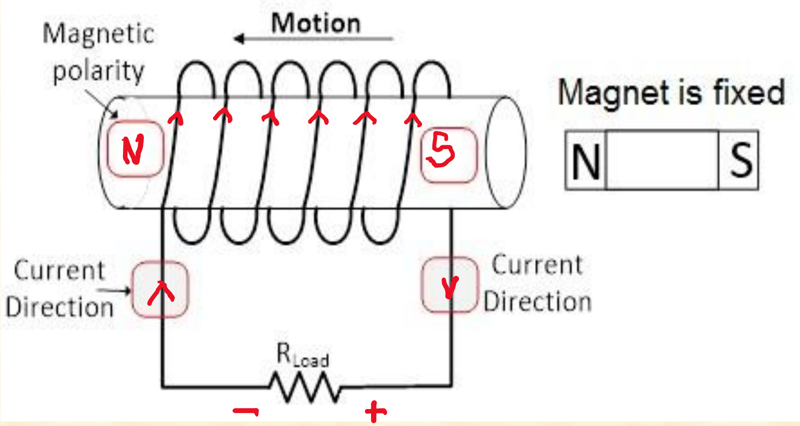
Number	Scientific Notation	Engineering Notation	Engineering Prefix Notation
0.046577	4.6577×10^{-2}	0.46577×10^{-3}	0.46577 m
0.0000578321	5.78321×10^{-5}	57.8321×10^{-6}	57.8321 µ
349.8	3.498×10^{2}	0.3498×10^{3}	0.3498 K
54945256.8	5.49452568×10^{7}	54.9452568×10^{6}	54.9452568 M

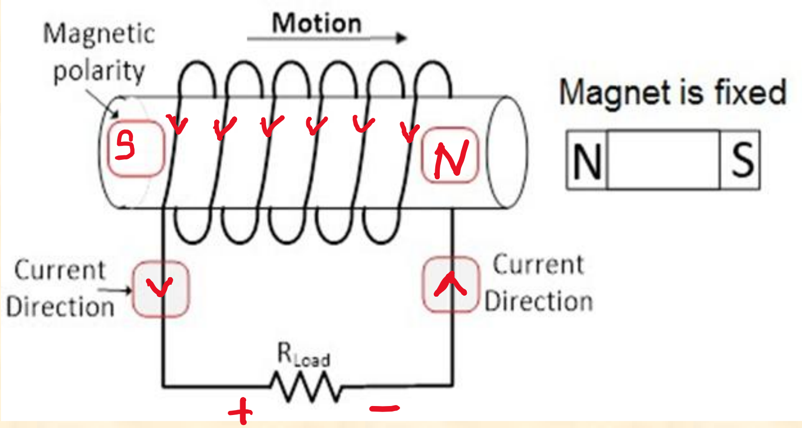
SI Units

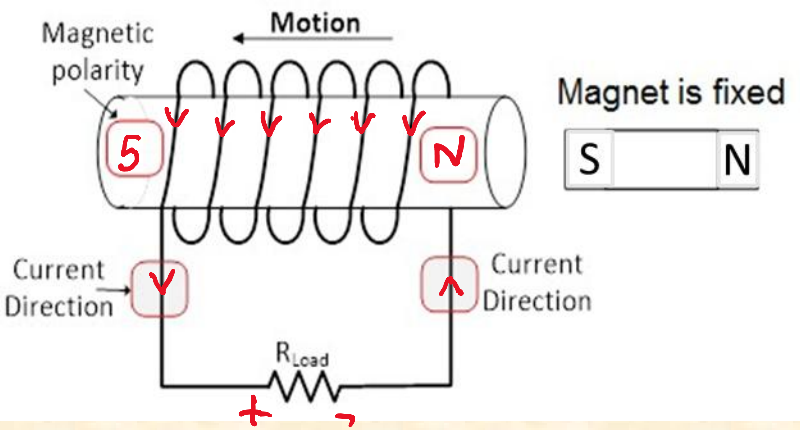
Parameter	Symbol	SI Unit
Voltage	V	Volt (V)
Current	I	Ampere (A)
Resistance	R	Ohm (Ω)
Power	P	Watts (W)
Capacitance	C	Farad (F)
Inductance	L	Henry (H)
Impedance	Z	Ohm (Ω)
Charge	Q	Coulomb (C)
Frequency	f	Hertz (Hz)
Period	T	Second (s)
Angular Frequency	ω	Rad/s

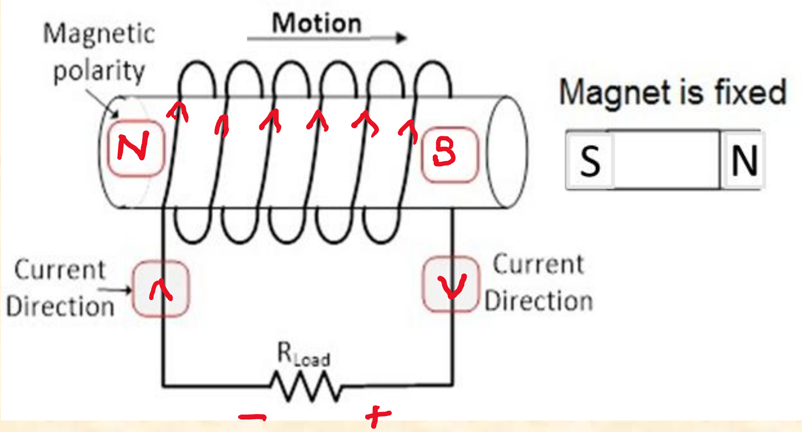
Parameter	Symbol	SI Unit
Magnetic Flux	Φ	Weber (Wb)
Magnetic Flux Density	B	Tesla (T) or Wb/m^2
Magnetomotive Force	mmf	$Ampere\ Turns\ (At)$
Reluctance	R	(At/Wb)
Permeability of free space	μ_0	$Wb/(At.m)$
Relative Permeability	μ_r	No unit
Permeability of material	$\mu_m=\mu_0\mu_r$	$Wb/(At.m)$
Permittivity of free space	ε_0	Farad per meter (F/m)
Relative permittivity	ε_r	No Unit
Time Constant	τ	Second (s)

Electromagnetic Induction

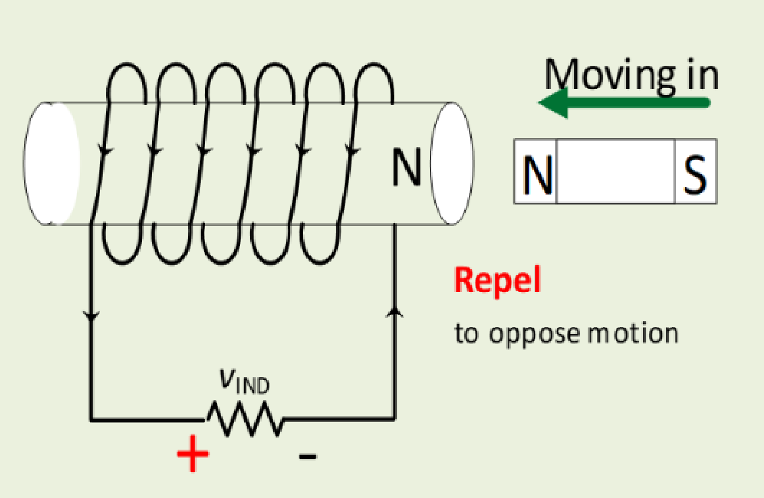


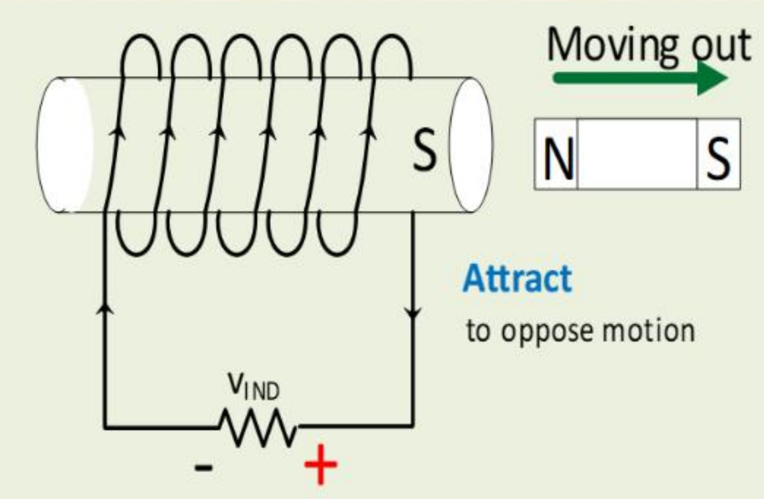


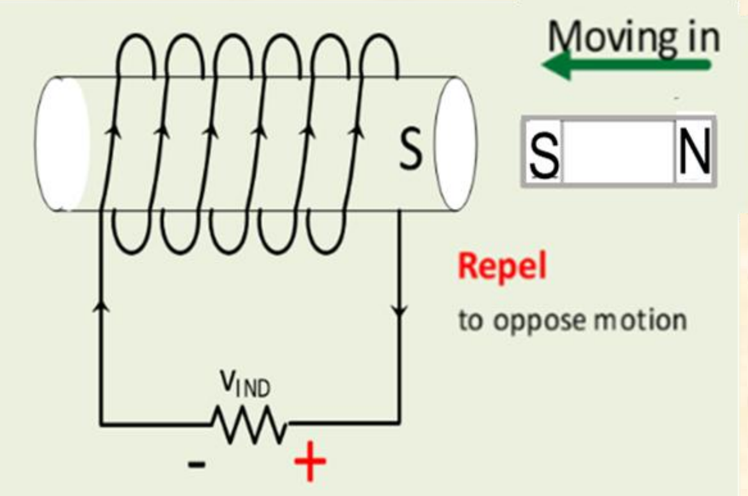


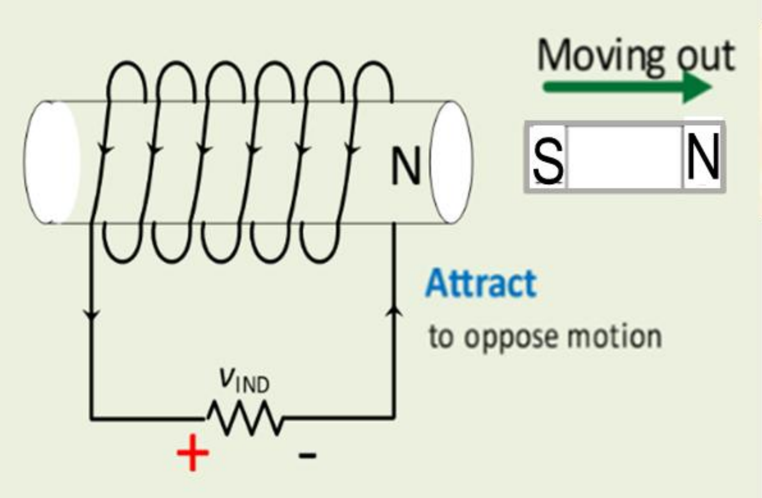


Electromagnetic Induction









Superposition Theorem

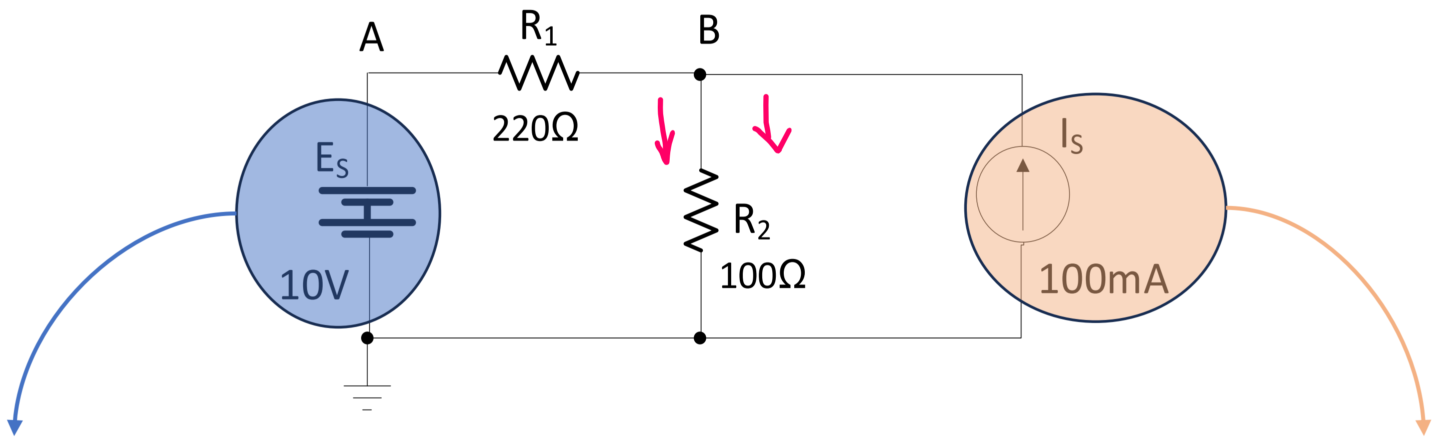
- The analysis of a circuit with **more than one source** can be determined by examining the **individual** contribution from each **source**.

- **All sources except one** must be **“turned off”** and evaluate the circuit currents and voltages.

- The **Final Step** is to determine the total voltage(or current) in any part of a circuit is **equal to the algebraic sum** of the voltages(or currents) produced by each source.

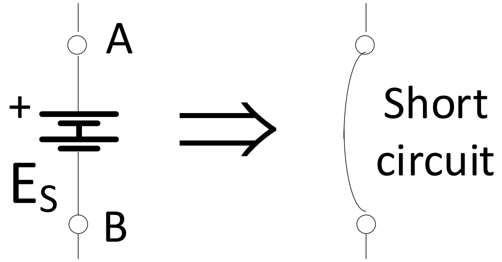
How to “turned off” the voltage & current sources

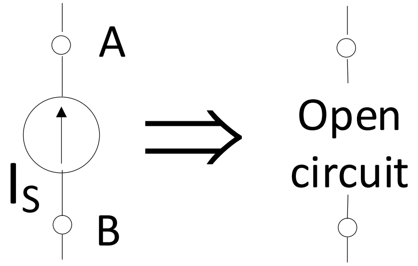
turned off ➡ **make zero voltage/current source**



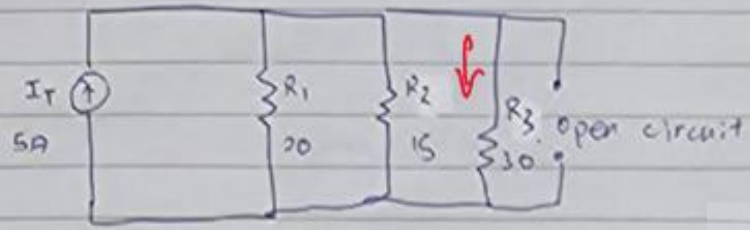
$E_S = V_{AB} = 0V$

$I_S = 0A$





Q2a)



$$R_{EQ} = \frac{1}{\frac{1}{20}+\frac{1}{15}+\frac{1}{30}}$$

$$= 6.67\Omega$$

$$I_3' = \frac{6.67}{30} \times 5$$

$$= 1.11A \text{ (2dp) } (\cancel{\uparrow}) \; (\downarrow)$$

current direction
$(\uparrow)$ or $(\downarrow)$ or $(\rightarrow)$ or $(\leftarrow)$

voltage polarity
$\left(\begin{smallmatrix}+\\-\end{smallmatrix}\right)$ $\left(\begin{smallmatrix}-\\+\end{smallmatrix}\right)$ $(+\ -)$ $(-\ +)$
+ on top + below + on left + on right

Q2b)

$$I_3'' = \frac{6.67}{30} \times 10$$

$$= 2.22A \text{ (2dp) } (\cancel{\uparrow}) \; (\uparrow)$$

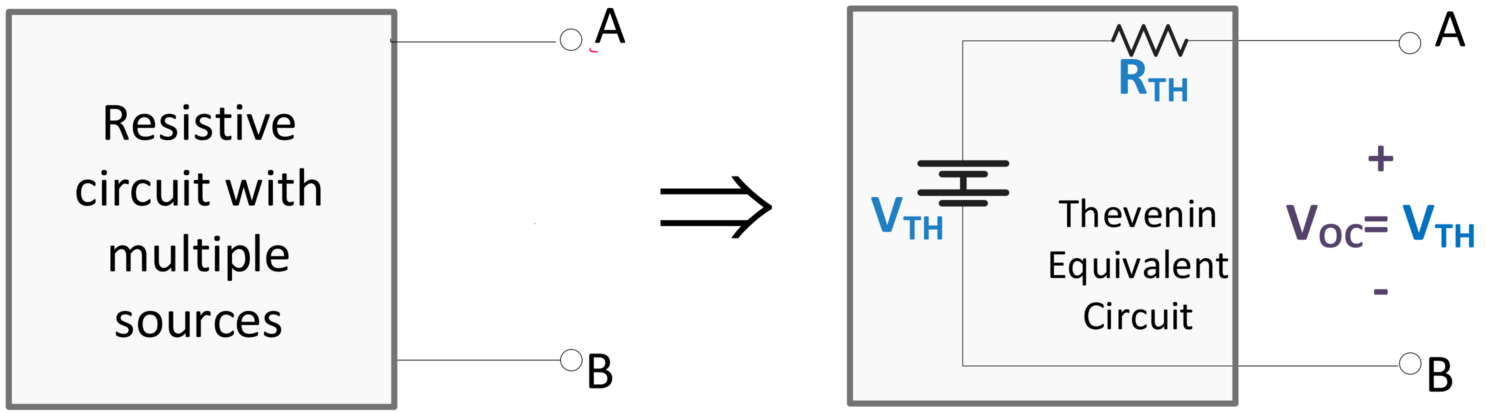
3a) $I_3 = I_3'' - I_3'$

$= 2.22 - 1.11$

$= 1.11A \; (\uparrow)$

Thevenin's Theorem

Thevenin's theorem states that **any resistive circuit network, can be replaced by voltage source in series with a source resistance** when viewed from the output terminals A-B.

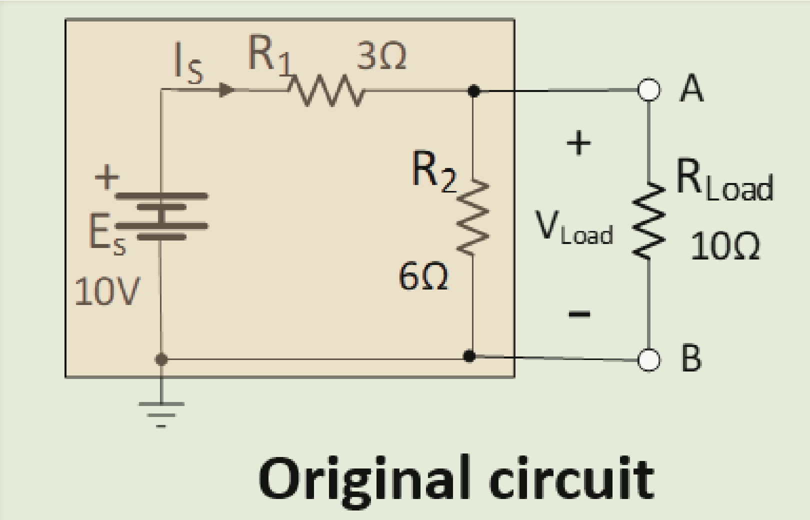


A Thevenin's Equivalent Circuit consists of:

- Voltage Source: **Thevenin Voltage, V_{TH}**
- Series Resistance: **Thevenin Resistance, R_{TH}**

- As there is an open circuit between terminal A-B, hence, voltage at the open terminal is known as **Open-Circuit Voltage (V_{OC})**

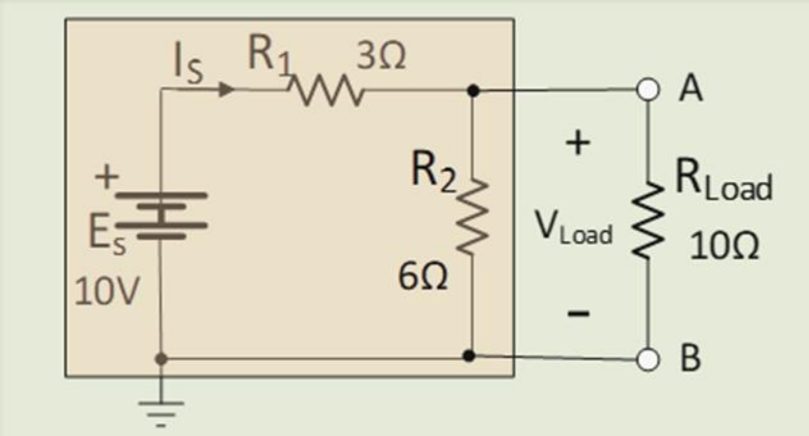
To find Thevenin's voltage, V_{TH}



Step 1 : Remove any load/resistor between terminal A-B

Step 2 : Use any appropriate circuit analysis method to find voltage value of VOC

To find Thevenin's Resistance, R_{TH}



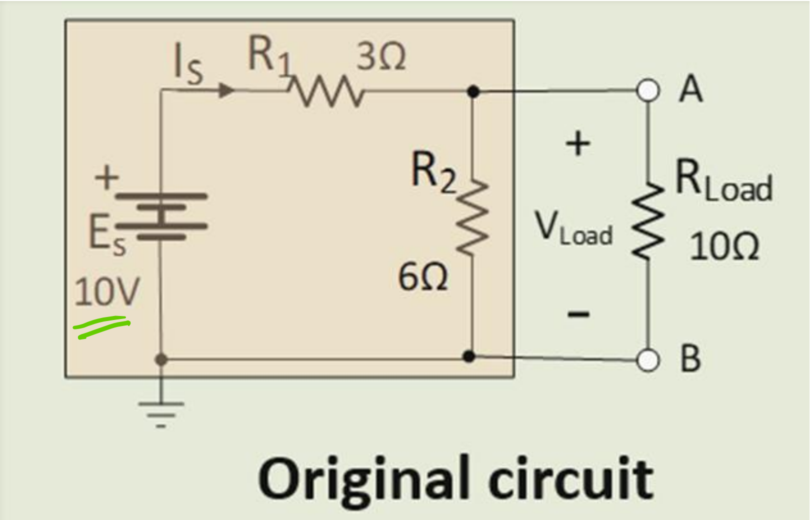
Original circuit

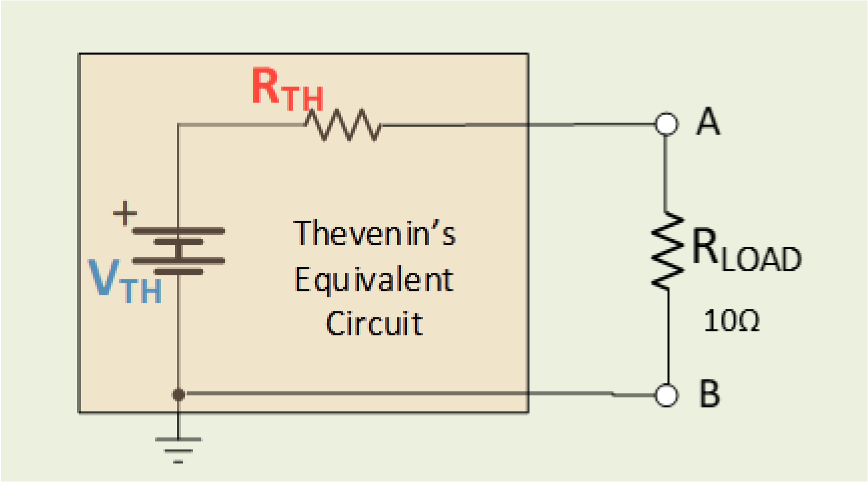
Step 1 : Remove any load/resistor between terminal A-B

Step 2 : Turn off (zero) all the sources (similar methods used in Superposition Theorem)

Step 3 : Find the equivalent resistance, REQ across the terminal A-B

To draw Thevenin's Equivalent Circuit

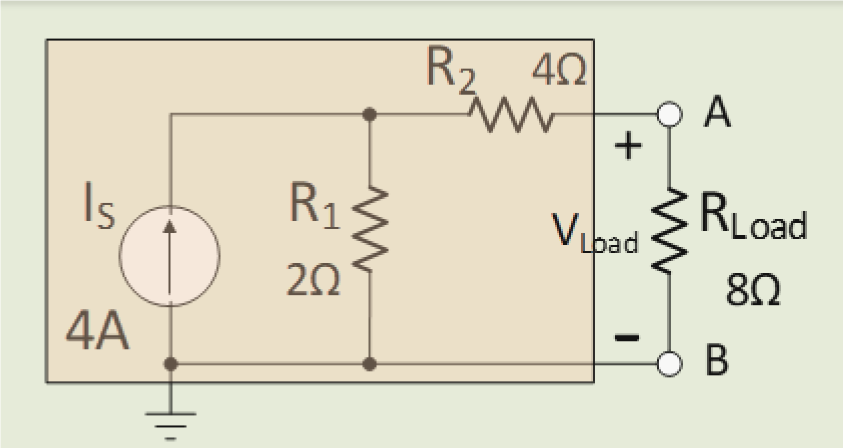




Step 1 : Draw the Thevenin's Equivalent Circuit with the load, **R_{LOAD}**, **re-connected** to terminal A-B

Step 2 : Analyse voltage and current for the load resistor, **R_{LOAD}**, when connected back to Thevenin's Equivalent Circuit.

To find Thevenin's voltage, V_{TH}

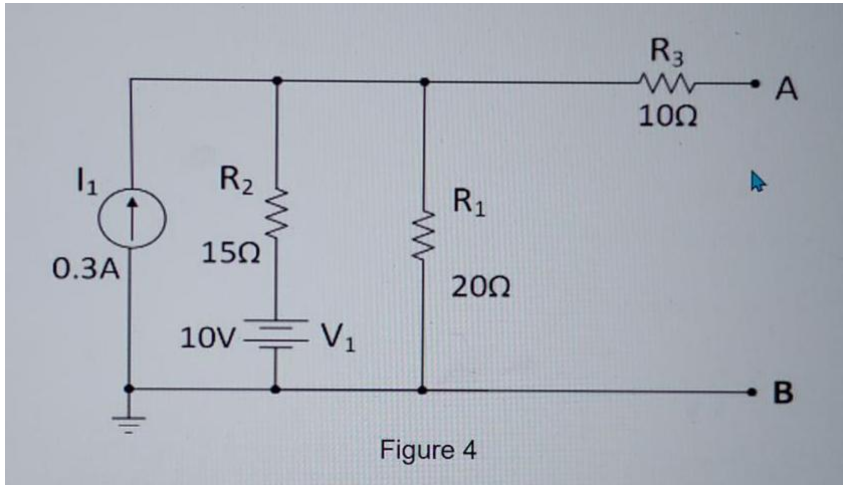


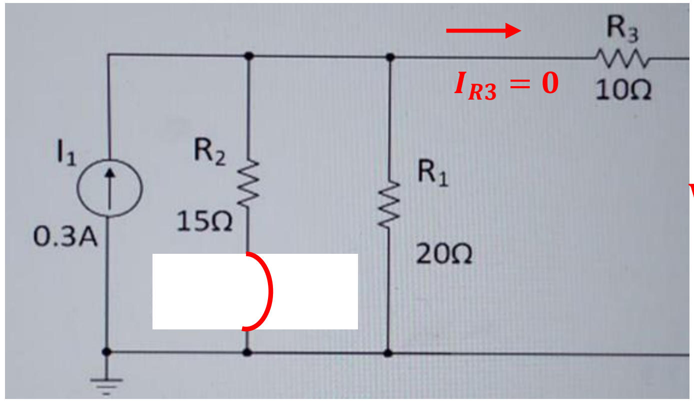
Original circuit

Step 1 : Remove any load/resistor between terminal A-B

Step 2 : Use any appropriate circuit analysis method to find voltage value of VOC

Apply Superposition Theorem to determine **Thevenin Equivalent Voltage, V_{TH}** & **Thevenin Equivalent Resistance R_{TH}** between node A & B.

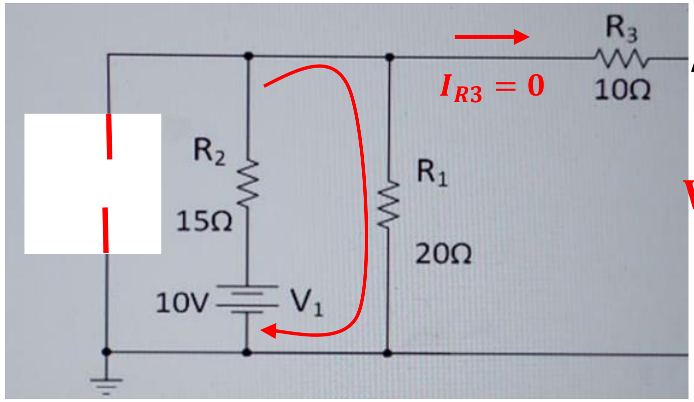




A

$V'_{TH} = V_{AB}$

B



A

$V''_{TH} = V_{AB} = V_{R1}$

B

I_1 splits into R_2 and R_1. Hence use current divider.

$$I_{R1} = \frac{R_{EQ}}{R_1} \times I_1 \rightarrow R_{EQ} = R_{12}$$

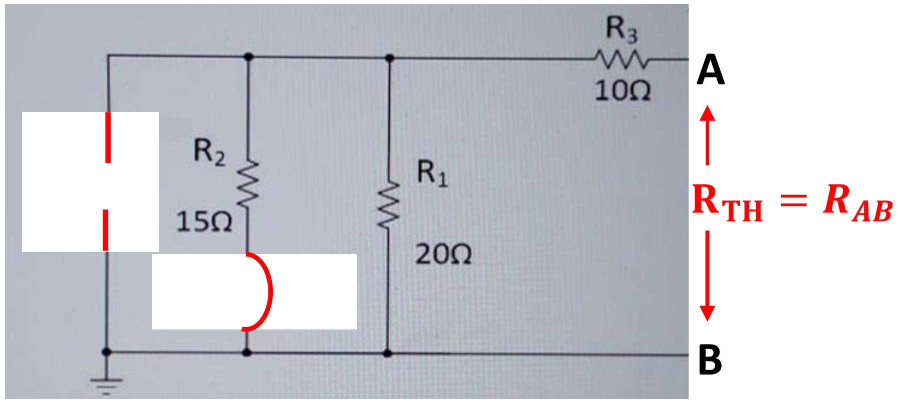
$$V'_{TH} = V_{AB} = V_{R1} + \cancelto{0}{V_{R3}} = I_{R1} \times R_1$$

Series circuit, current the same. Hence use voltage divider.

$$V_{R1} = \frac{R_1}{R_T} \times V_1$$

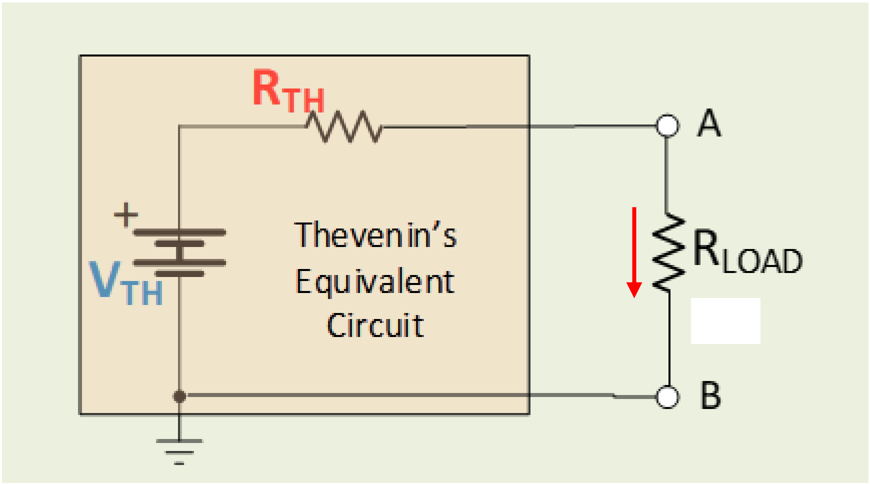
$$V_{R1} V_{AB} = V_{R1} + \cancelto{0}{V_{R3}} = V_{R1}$$

$$V_{TH} = V'_{TH} + V''_{TH}$$



To find R_{TH}, all sources must be turned "OFF"

$$R_{TH} = R_3 + R_1 \,//\, R_2$$



$$V_{RLoad} = \frac{R_{Load}}{R_{Total}} \times V_{TH}$$

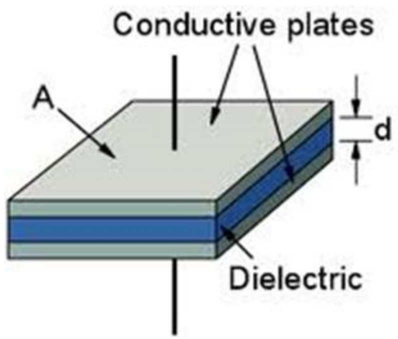
$$I_{Load} = \frac{V_{Rload}}{R_{Load}}$$

Capacitor

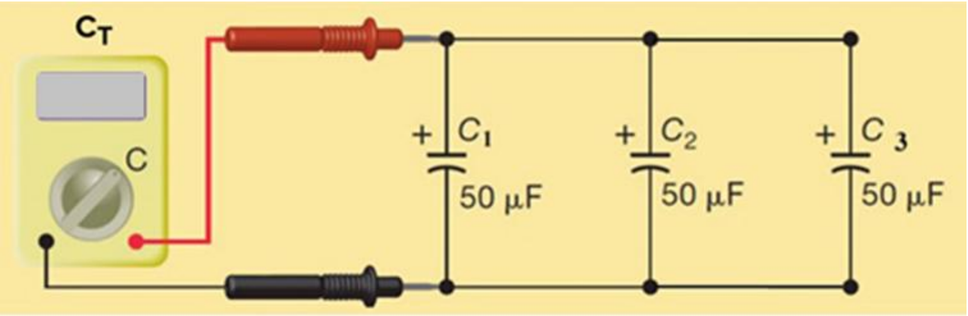
- The capacitance of a parallel-plate capacitor is defined by the equation,

$$C = \varepsilon_0 \varepsilon_r \frac{A}{d} \quad (F)$$

ε_0 = *permittivity of free space ≈ 8.85 x 10^{-12} F/m*
ε_r = *relative permittivity of the dielectric material*
A = area of the plates, m²
d = distance between 2 plates



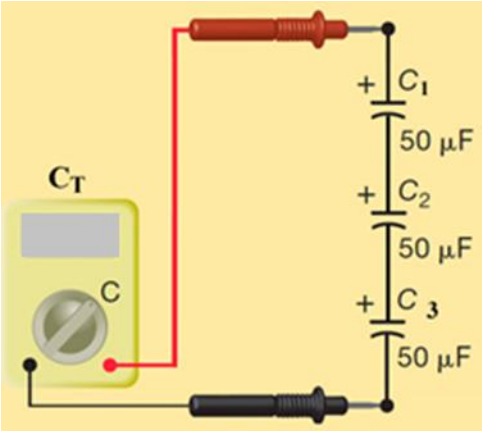
Capacitor in parallel



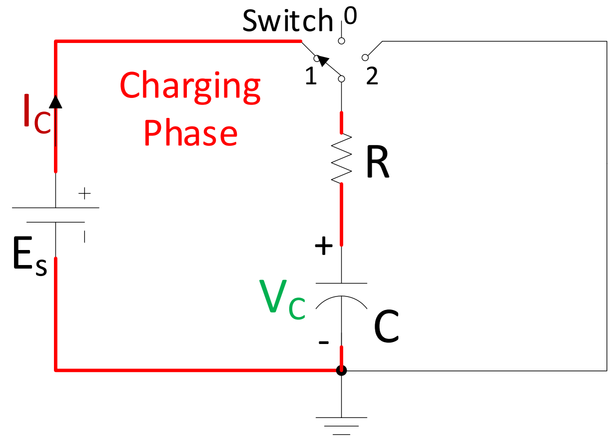
$$C_T = C_1 + C_2 + \dots + C_n$$

Capacitor in series

$$C_T = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

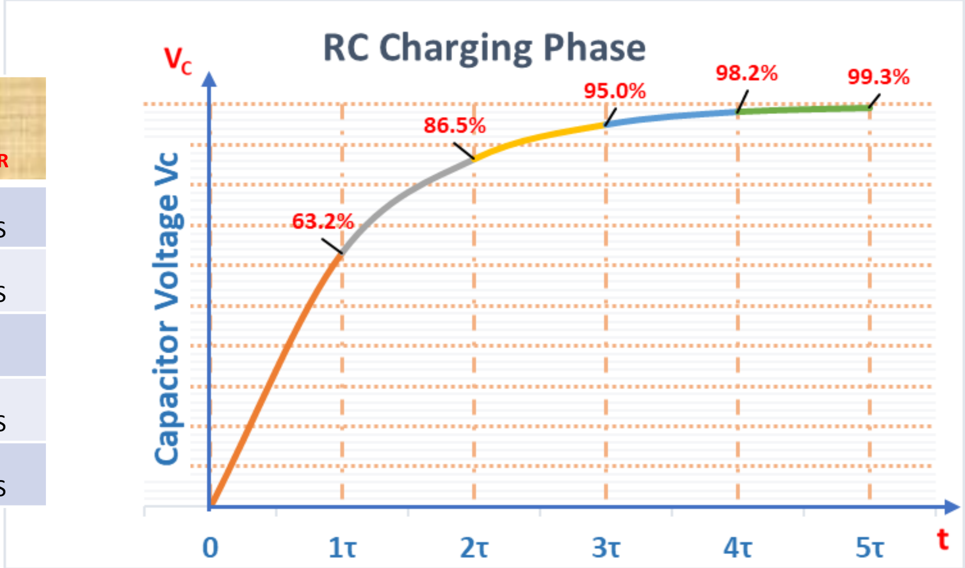


Transient curves for series RC circuits - Charging phase



The Time Constant $(\tau) = R * C$

Time Constant τ	Capacitor Voltage V_C	Resistor Voltage V_R
1τ	$0.632*E_S$	$0.368*E_S$
2τ	$0.865*E_S$	$0.135*E_S$
3τ	$0.95 * E_S$	$0.05*E_S$
4τ	$0.982*E_S$	$0.018*E_S$
5τ	$0.993*E_S$	$0.007*E_S$

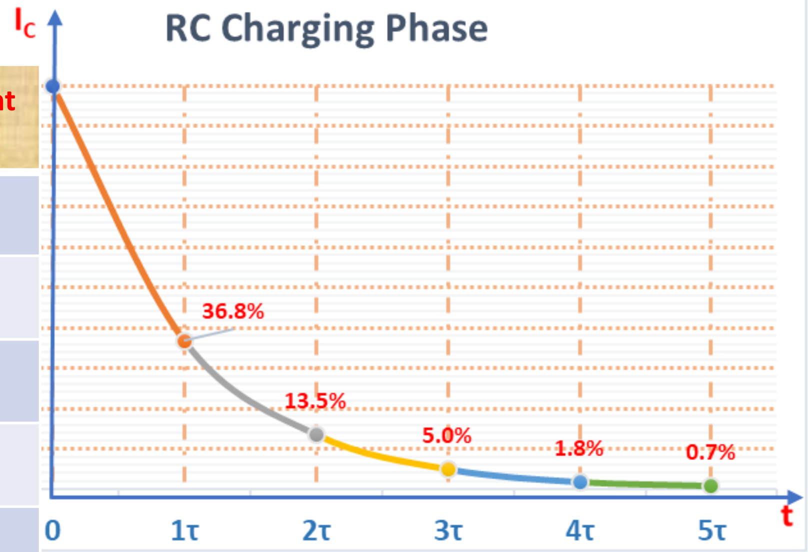


Capacitor Voltage V_C

Initial Value = 0 V

Final steady Value = E_S

Time Constant τ	Capacitor Current I_C
1τ	$0.368 * \frac{E_S}{R}$
2τ	$0.135 * \frac{E_S}{R}$
3τ	$0.05 * \frac{E_S}{R}$
4τ	$0.018 * \frac{E_S}{R}$
5τ	$0.007 * \frac{E_S}{R}$



Capacitor Current I_C

Initial Value = $\frac{E_S}{R}$

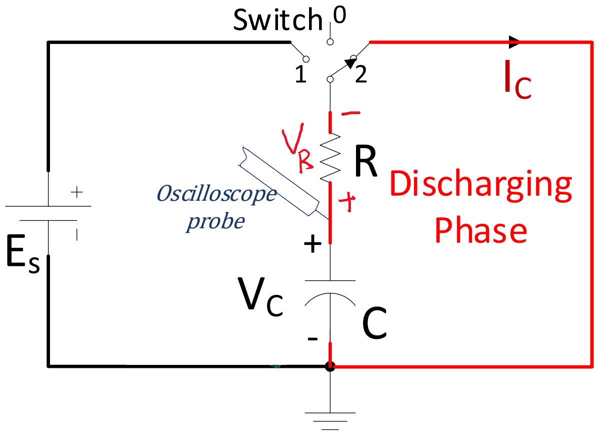
Final steady Value = 0 A

$Note : i_C(t) = C\frac{dv}{dt}$,

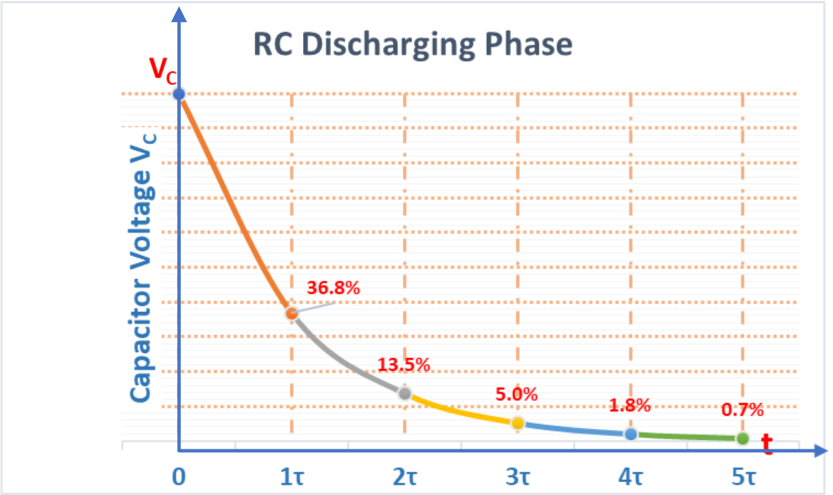
i_C is max when $(\frac{dv}{dt})$ is max, i.e V_C = 0 at t=0

- It takes approximately 5τ to reach its final steady-state value

Transient curves for series RC circuits - Discharging phase



Time Constant τ	Capacitor Voltage V_C	Resistor Voltage V_R
1τ	$0.368*E_S$	$0.368*E_S$
2τ	$0.135*E_S$	$0.135*E_S$
3τ	$0.05*E_S$	$0.05*E_S$
4τ	$0.018*E_S$	$0.018*E_S$
5τ	$0.007*E_S$	$0.007*E_S$



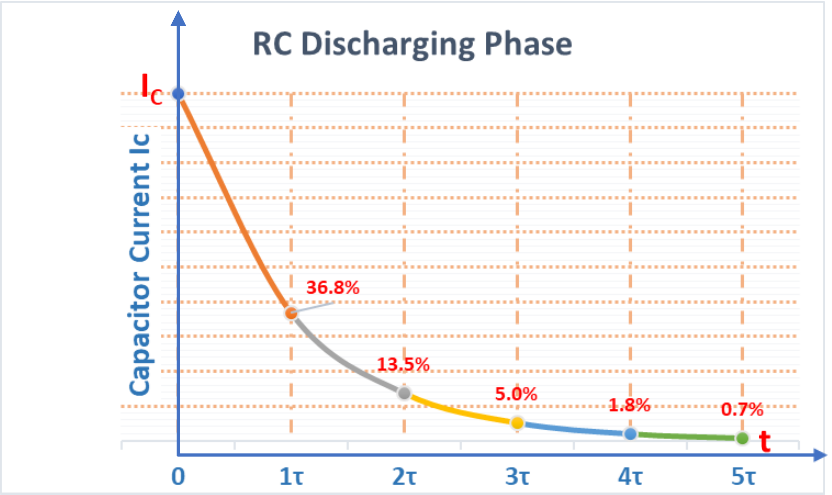
The Time Constant $(\tau) = R * C$

Capacitor Voltage V_C

Initial Value = E_S

Final steady Value = 0 V

Time Constant τ	Capacitor Current I_C
1τ	$0.368 * \frac{E_S}{R}$
2τ	$0.135 * \frac{E_S}{R}$
3τ	$0.05 * \frac{E_S}{R}$
4τ	$0.018 * \frac{E_S}{R}$
5τ	$0.007 * \frac{E_S}{R}$



Capacitor Current I_C

Initial Value = $\frac{E_S}{R}$

Final steady Value = 0 A

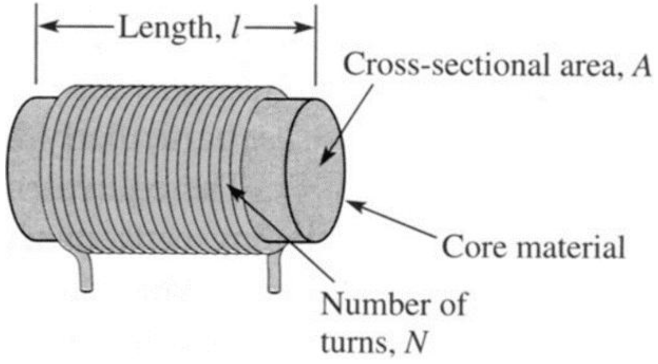
- It takes approximately 5τ to reach its final steady-state value

Inductor

- The inductance of a coil can be approximated from its physical characteristics as

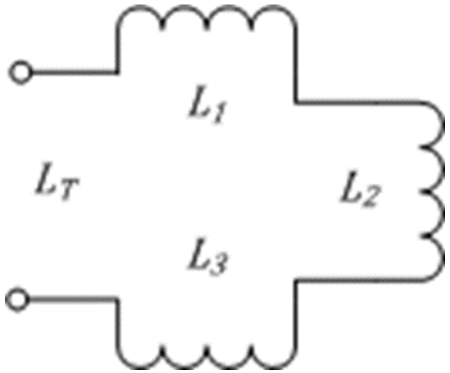
$$L = \mu_m \frac{N^2 A}{l} \ (\text{H})$$

μ_m = permeability of the core, Wb/At*m
A = cross-sectional area of the core, m^2
l = length of the coil, m



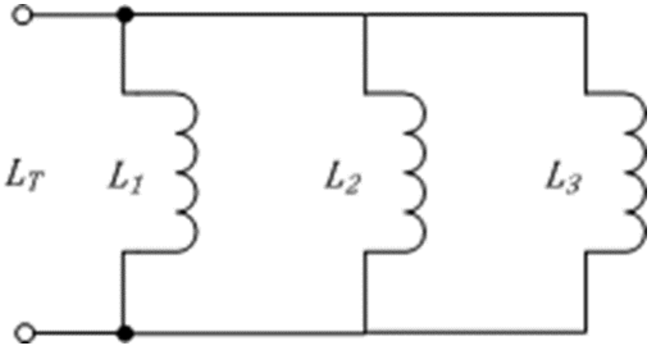
Inductors in series

$$L_T = L_1 + L_2 + L_3$$

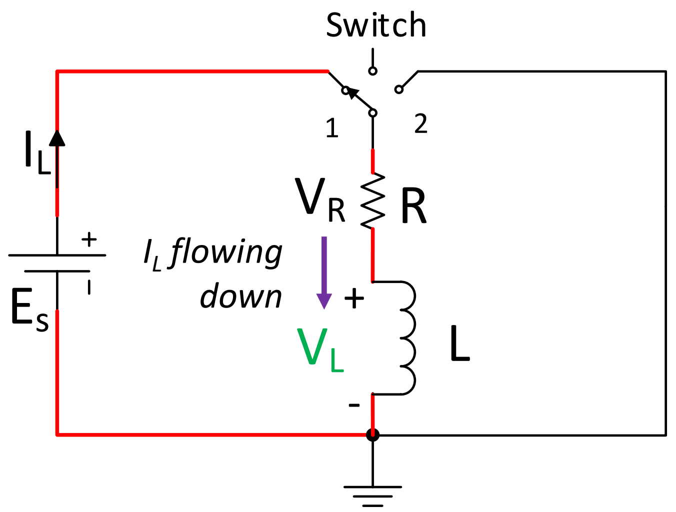


Inductors in parallel

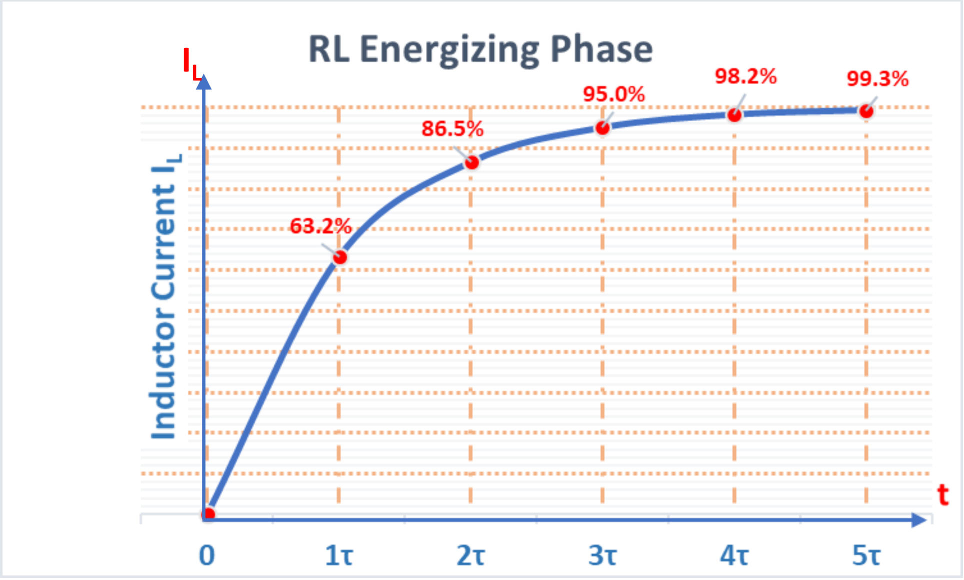
$$L_{EQ} = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}}$$



Transient curves for series RL circuits - Energizing phase



Time Constant τ	Inductor Current I_L
1τ	$0.632 * \frac{E_S}{R}$
2τ	$0.865 * \frac{E_S}{R}$
3τ	$0.95 * \frac{E_S}{R}$
4τ	$0.982 * \frac{E_S}{R}$
5τ	$0.993 * \frac{E_S}{R}$



The Time Constant $(\tau) = \frac{L}{R}$

Inductor Current I_L

Initial Value = 0 A

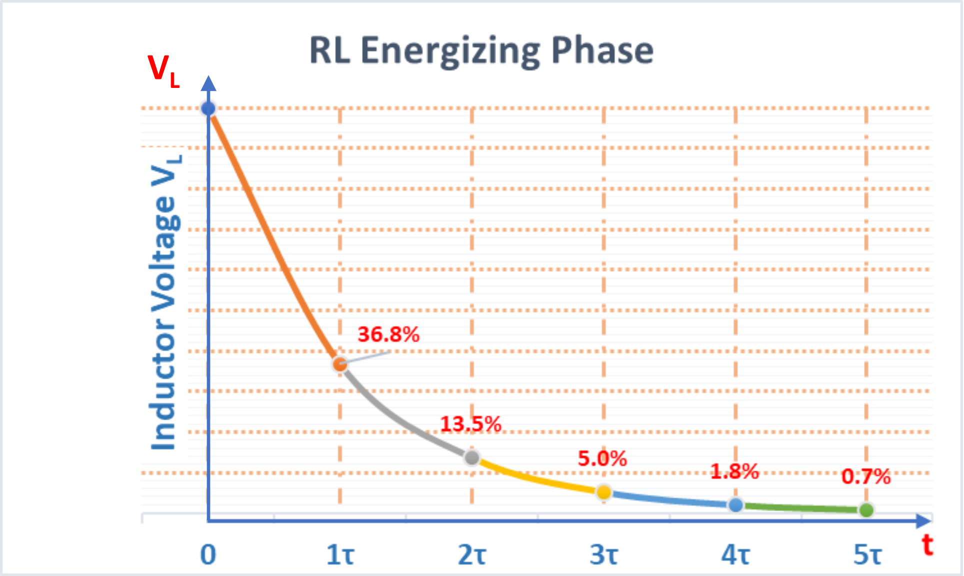
Final steady Value = $\frac{E_S}{R}$

Inductor Voltage V_L

Initial Value = E_S

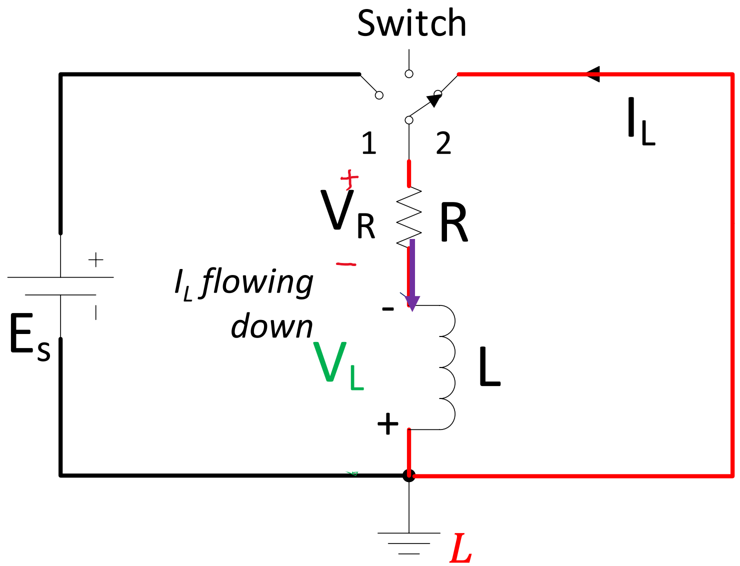
Final steady Value = 0 V

Time Constant τ	Inductor Voltage V_L	Resistor Voltage V_R
1τ	$0.368*E_S$	$0.632*E_S$
2τ	$0.135*E_S$	$0.865*E_S$
3τ	$0.05*E_S$	$0.95 * E_S$
4τ	$0.018*E_S$	$0.982*E_S$
5τ	$0.007*E_S$	$0.993*E_S$



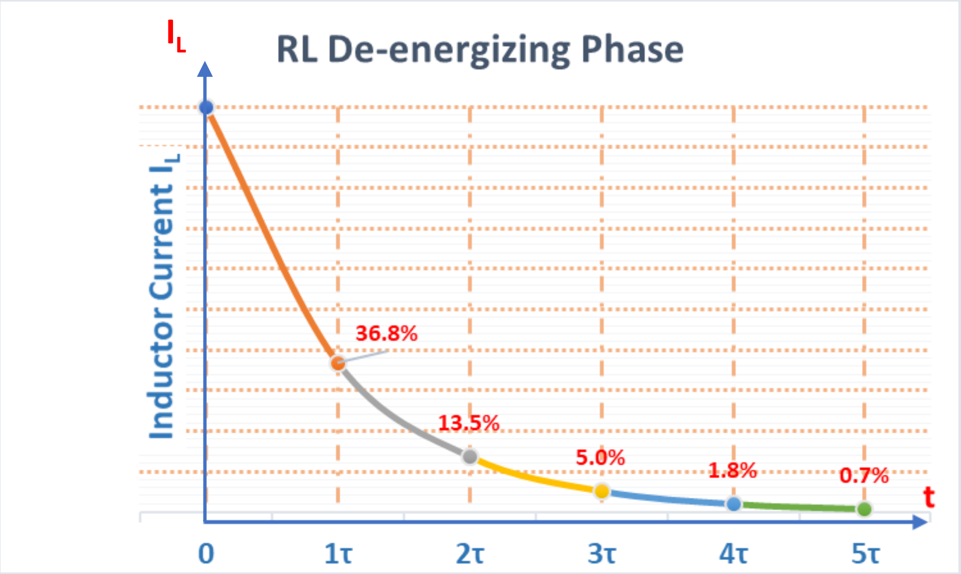
- It takes approximately 5τ to reach its final steady-state value

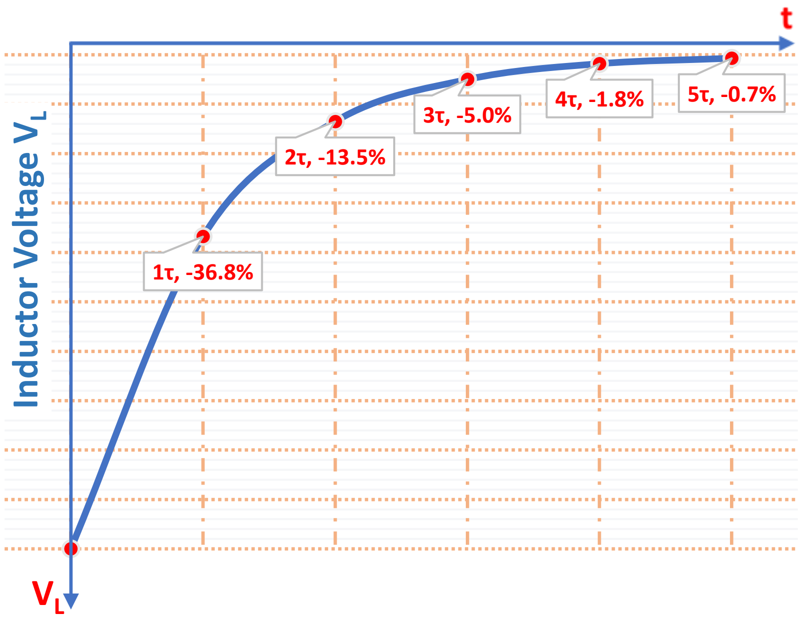
Transient curves for series RL circuits – De-energizing phase



The Time Constant (τ) $= \dfrac{L}{R}$

Time Constant τ	Inductor Current I_L
1τ	$0.368 * \frac{E_S}{R}$
2τ	$0.135 * \frac{E_S}{R}$
3τ	$0.05 * \frac{E_S}{R}$
4τ	$0.018 * \frac{E_S}{R}$
5τ	$0.007 * \frac{E_S}{R}$





Inductor Current I_L

Initial Value $= \frac{E_S}{R}$

Final steady Value = 0 A

Inductor Voltage V_L

Initial Value = - E_S

Final steady Value = 0 V

Time Constant τ	Inductor Voltage V_L	Resistor Voltage V_R
1τ	- 0.368*E_S	0.368*E_S
2τ	- 0.135*E_S	0.135*E_S
3τ	- 0.05*E_S	0.05*E_S
4τ	- 0.018*E_S	0.018*E_S
5τ	- 0.007*E_S	0.007*E_S

Note : The **inverted V_L**, also term as **"Back EMF"** reduces its strength as the magnetic energy collapses gradually.
The current falls exponentially to zero with the decay in the magnetic field

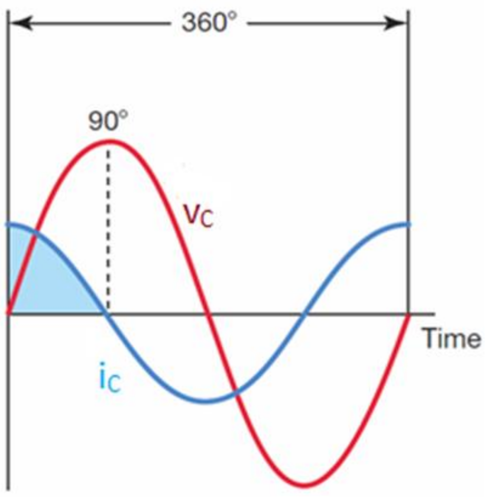
Capacitor - C

Capacitive Reactance – X_C

➢ $X_C(\Omega) = \frac{1}{2\pi f c}$

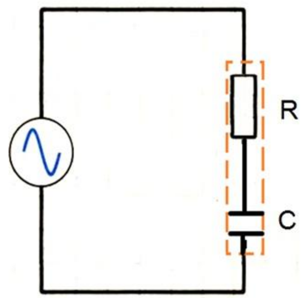
❖ The **current i_c leads the voltage** V_c in the ideal **capacitor** by **90^o**

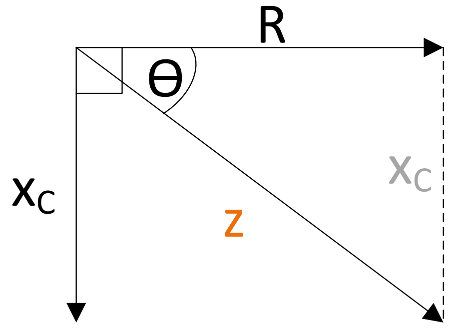
❖ Current through X_C, $i_C = v_C / x_C$



Current – Capacitor - Voltage

Impedance – z (Ω)

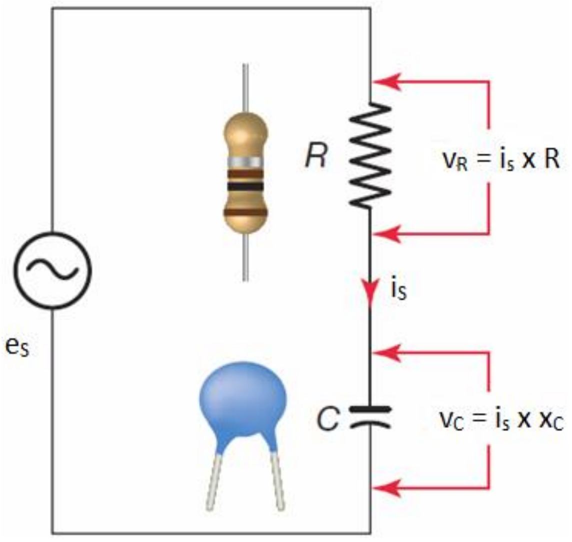




Formula for Impedance, z (Ω):

- Magnitude of z (Ω): $|Z| = \sqrt{R^2 + {x_C}^2}$
- z = $\sqrt{R^2 + {x_C}^2} \angle \theta° \; (\Omega)$
- $\theta° = \tan^{-1}\left(-\frac{x_C}{R}\right)$

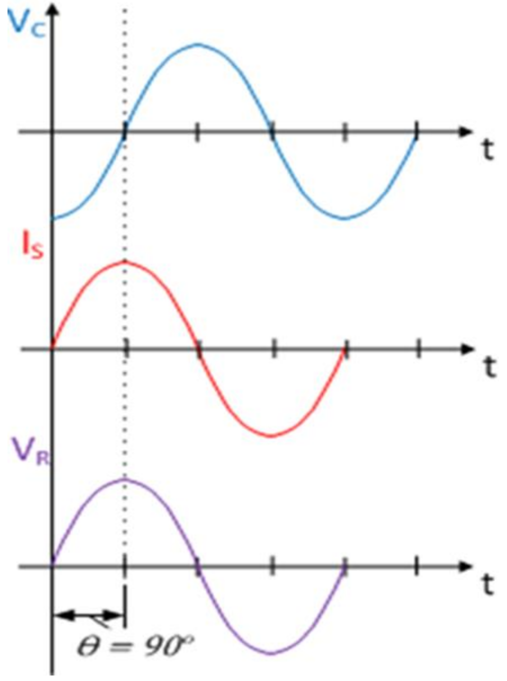
Series RC Circuit



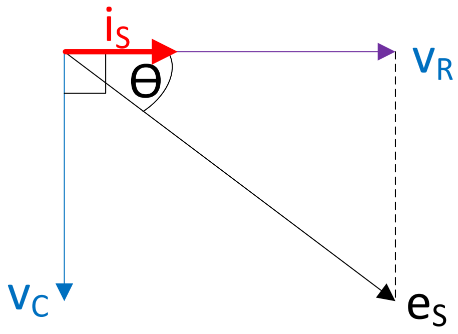
$$i_S = \frac{e_S}{Z}$$

$$V_R = i_S \times R$$

$$V_C = i_S \times X_C$$



Voltage phasor diagram



- For Capacitor → Current, i_s leads voltage, v_c by **90°**
- For Resistor → Current, i_s is always in phase with voltage V_R.

$$\boldsymbol{e_S} = \sqrt{v_R^2 + v_C^2} \angle \theta° \text{ (V)}$$

$$\text{where } \theta° = \tan^{-1}\left(-\frac{x_C}{R}\right)$$

EXAMPLE 17-14

Problem:

For the series *RC* circuit shown in **Figure 17-29**:

a. Calculate the value of the current flow.

b. Calculate the value of the voltage drop across the resistor.

c. Calculate the value of the voltage drop across the capacitor.

d. Calculate the circuit phase angle based on the voltage drops across the resistor and capacitor.

e. Express all voltages in polar notation.

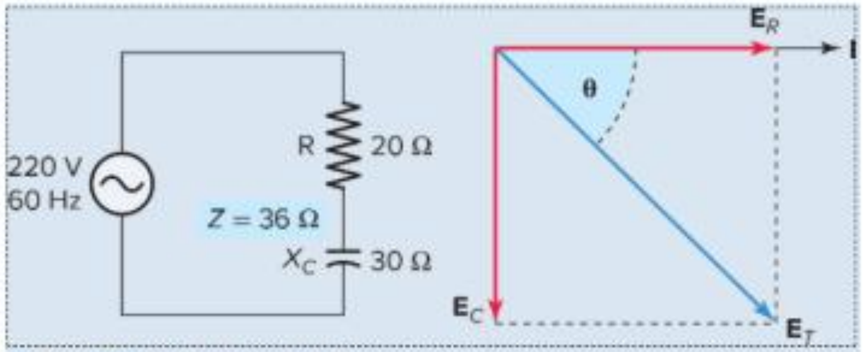


Figure 17-29 Circuit for example 17-14.

Solution:

a. $I = \dfrac{E_T}{Z}$

$= \dfrac{220\ \text{V}}{36\ \Omega}$

$= 6.11\ \text{A}$

b. $E_R = I \times R$

$= 6.11\ \text{A} \times 20\ \Omega$

$= 122\ \text{V}$

c. $E_C = I \times X_C$

$= 6.11\ \text{A} \times 30\ \Omega$

$= 183\ \text{V}$

d. $\theta = \tan^{-1} \dfrac{E_C}{E_R}$

$= \tan^{-1} \dfrac{183}{122}$

$= \tan^{-1}(1.5)$

$= 56.31°$

The calculated circuit phase angle is the same as that determined in the previous example using the ratio of X_C /R.

e. $E_T = 220\ \text{V} \angle -56.31°$

$E_R = 122\ \text{V} \angle 0°$

$E_C = 183\ \text{V}\ \ -90°$

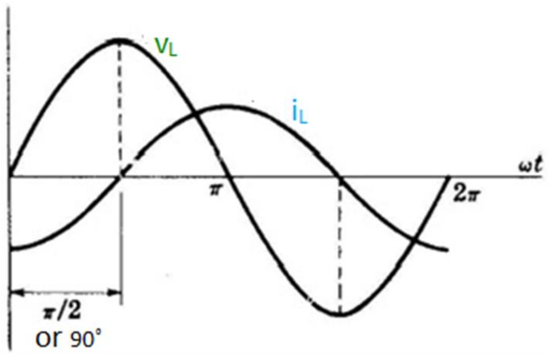
Inductor – L

Inductive Reactance – X_L

➢ $X_L(\Omega) = 2\pi f L$

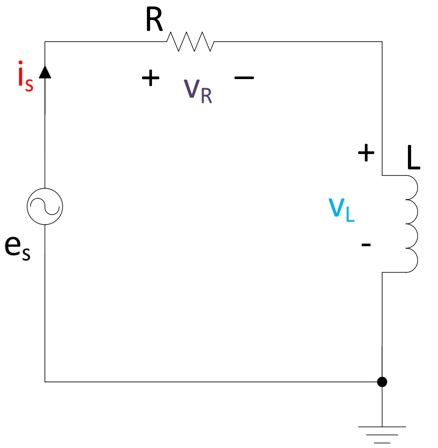
❖ In a purely inductive circuit, i_L **always lag the voltage** V_L by **90° or $\pi/2$.**

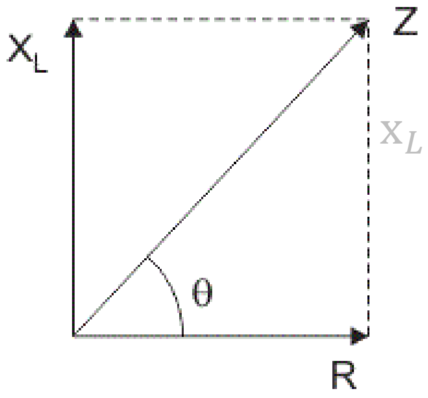
❖ Current through X_L, $i_L = v_L / x_L$



Voltage – Inductor - Current

Impedance – z (Ω)





Formula for Impedance, z (Ω):

- Magnitude of z (Ω): $|Z| = \sqrt{R^2 + x_L^2}$
- $z = \sqrt{R^2 + x_L^2} \angle \theta° \ (\Omega)$
- $\theta° = \tan^{-1}\left(\frac{x_L}{R}\right)$

EXAMPLE 17-9

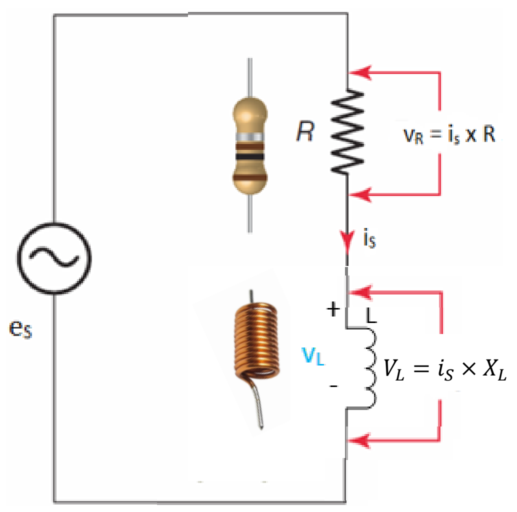
Problem:

An AC series *RL* circuit is made up of a resistor that has a resistance value of 150 Ω and an inductor that has an inductive reactance value of 100 Ω. Calculate the impedance and the phase angle theta (θ) of the circuit.

Solution:

$$\begin{aligned} Z &= \sqrt{R^2 + X_L^2} \\ &= \sqrt{150^2 + 100^2} \\ &= \sqrt{32{,}500} \\ &= 180\ \Omega \\ \theta &= \tan^{-1}\left(\frac{X_L}{R}\right) \\ &= \tan^{-1}\left(\frac{100}{150}\right) \\ &= \tan^{-1}\ (0.667) \\ &= 33.7^\circ \end{aligned}$$

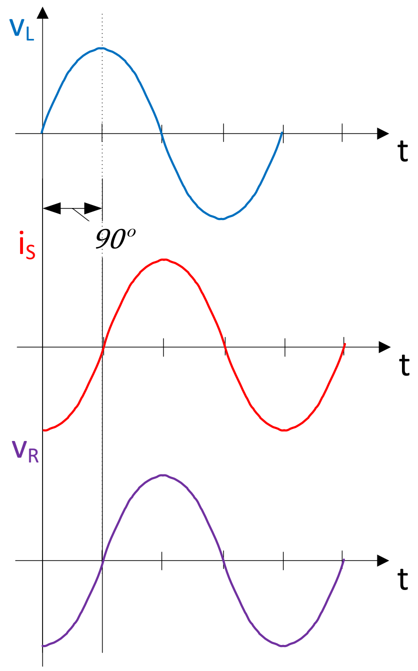
Series RL Circuit



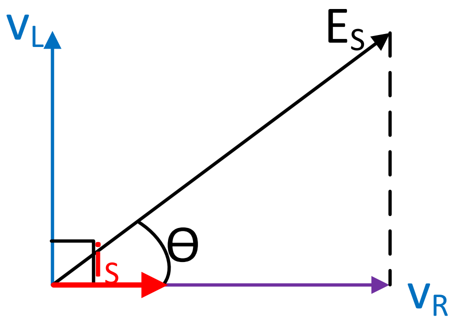
$$i_S = \frac{e_S}{Z}$$

$$V_R = i_S \times R$$

$$V_L = i_S \times X_L$$



Voltage phasor diagram



- For Inductor → Current, i_s lags voltage, v_L by **90°**
- For Resistor → Current, i_s is always in phase with voltage V_R.

$$e_S\ (V) = \sqrt{v_R^{\,2} + v_L^{\,2}} \angle \theta°$$

$$\text{where } \theta° = \tan^{-1}\left(\frac{x_L}{R}\right)$$

EXAMPLE 17-10

Problem:

For the series *RL* circuit shown in **Figure 17-17**:

a. Calculate the value of the current flow.

b. Calculate the value of the voltage drop across the resistor.

c. Calculate the value of the voltage drop across the inductor.

d. Calculate the circuit phase angle based on the voltage drops across the resistor and inductor.

e. Express all voltages in polar notation.

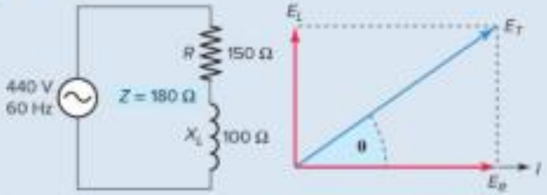


Figure 17-17 *RL* series circuit for example 17-10.

Solution:

a. $I = \frac{E_T}{Z}$

$= \frac{440\ \text{V}}{180\ \Omega}$

$= 2.44\ \text{A}$

b. $E_R = I \times R$

$= 2.44\ \text{A} \times 150\ \Omega$

$= 366\ \text{V}$

c. $E_L = I \times X_L$

$= 2.44\ \text{A} \times 100\ \Omega$

$= 244\ \text{V}$

d. $\theta = \tan^{-1} \frac{E_L}{E_R}$

$= \tan^{-1} \frac{244\ \text{V}}{366\ \text{V}}$

$= \tan^{-1}(0.667)$

$= 33.7°$

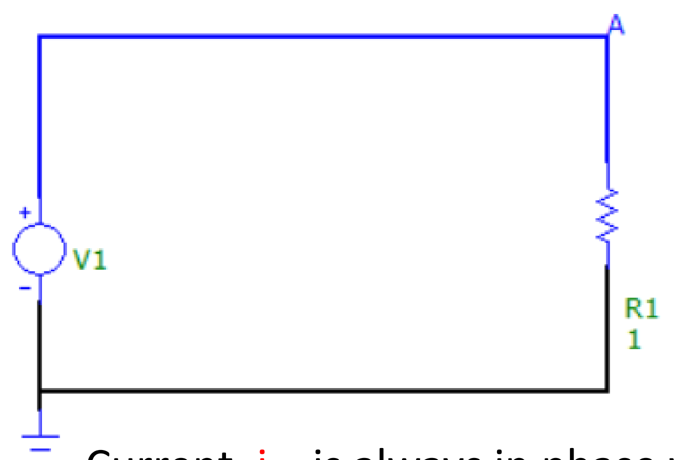
e. $E_T = 440\ \text{V} \angle 33.7°$

$E_R = 366\ \text{V} \angle 0°$

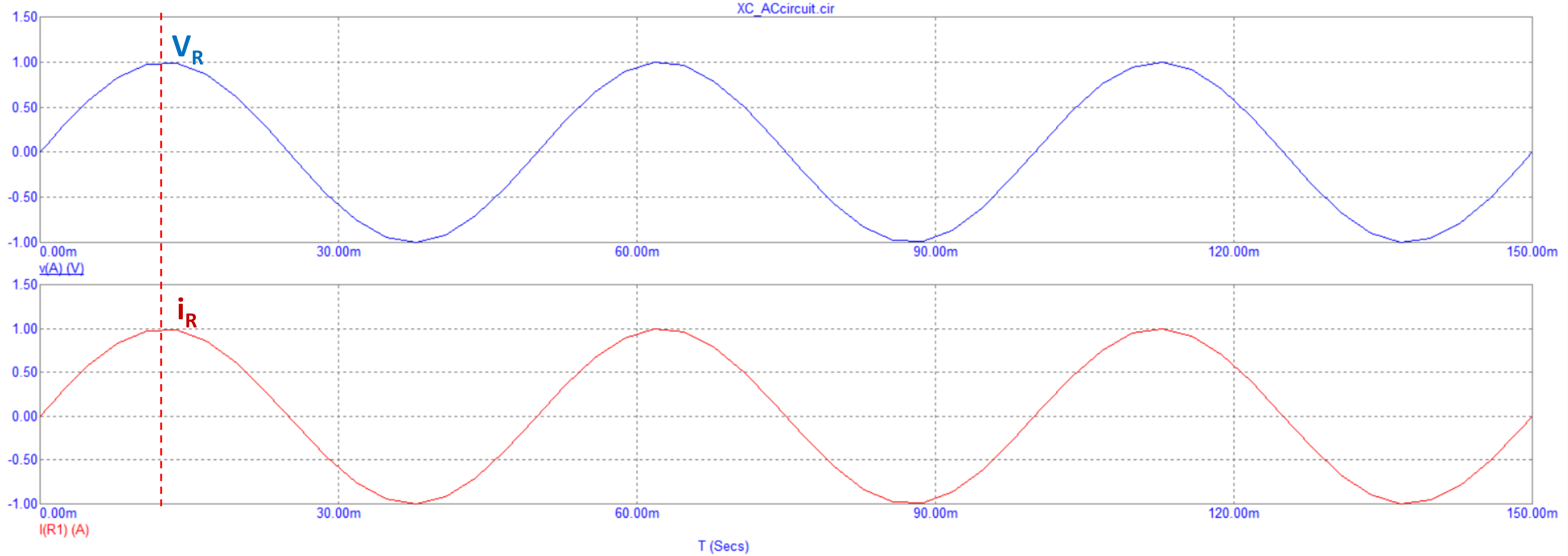
$E_L = 244\ \text{V} \angle 90°$

Question a) Describe the phase relationship between V_{R1} and I_{R1}

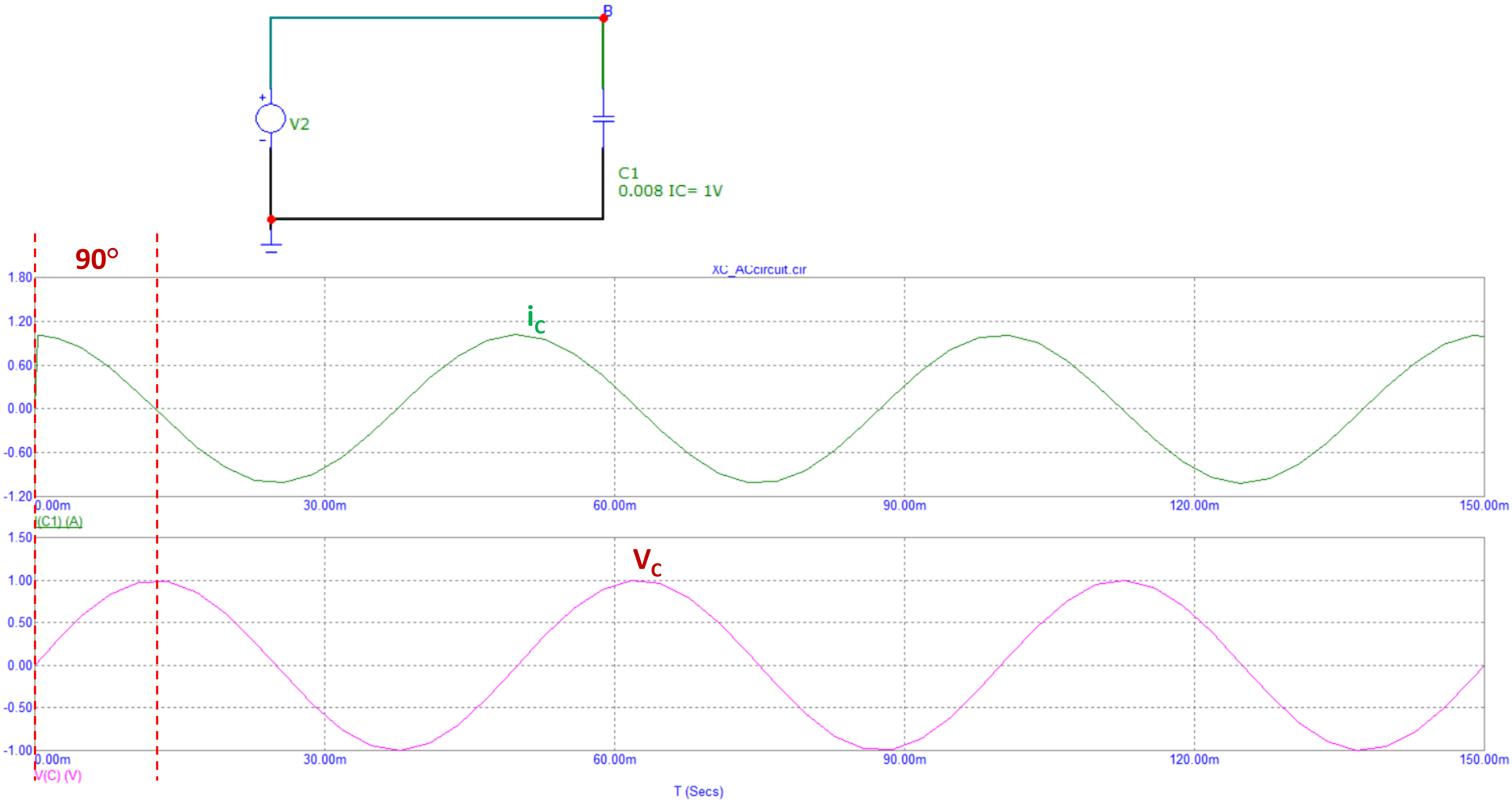
Q3



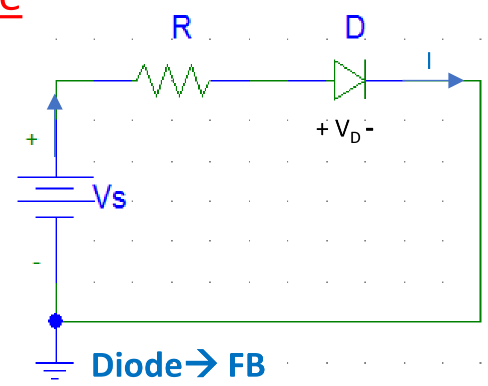
Current, i_{R1} is always in phase with voltage V_R.



Question b) Describe the phase relationship between V_{C1} and I_{C1}



Diode



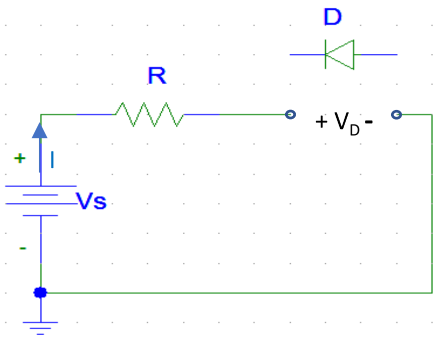
Diode→ FB

$V_D = V_t = (0.7V)$

$$I = \frac{V_s - V_D}{R}$$

$$Power\,(Diode) = I \;\; X \;\; V_D$$

$$Power\,(Resistor) = I^2 \;\; R$$



Diode→ RB → Diode is considered Open

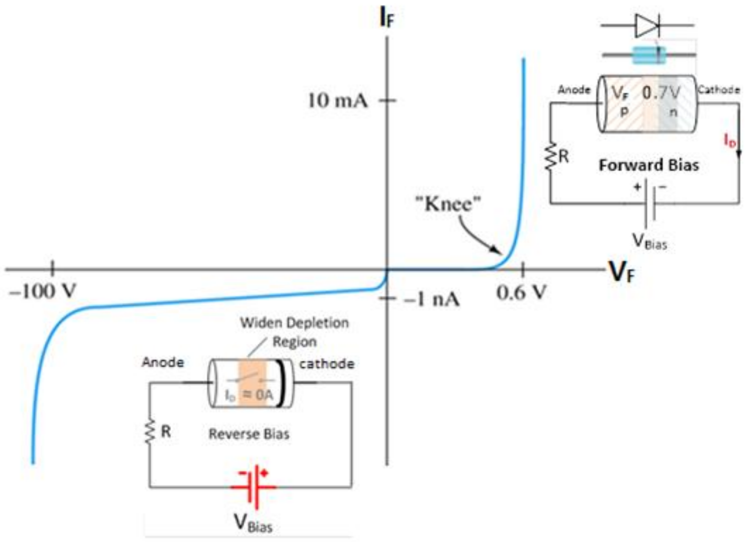
$V_D = V_s$

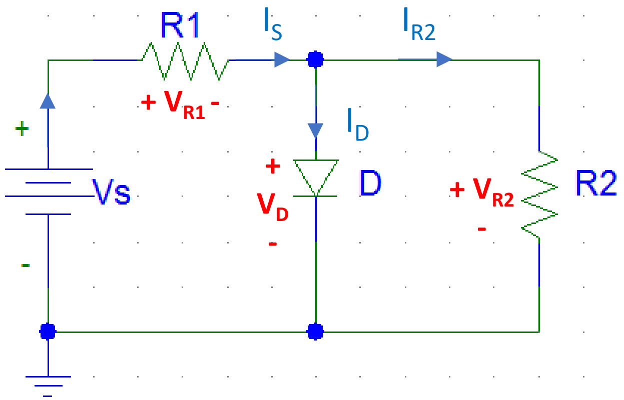
$$I = 0$$

Power dissipation for diode

P = VI only

Don’t use I^2 R or V^2 / R





Diode→ Forward Bias, $V_D = V_t = (0.7V)$

$$I_s = \frac{V_s - V_D}{R_1} \;, \; I_{R2} = \frac{V_D}{R_2}$$

$$I_D = I_s - I_{R2}$$

$$Power\,(Diode) = I_D \;\; X \;\; V_D$$

$$Power\,(Resistor) = {I_S}^2 \;\; R$$

Zener Diode

Zener Diodes (unlike normal Diodes) are operated in reverse biased condition and used for voltage regulation.
Forward Bias (Fig 1) → Voltage across Zener diode = V_D = 0.7V (Same as normal Diode)

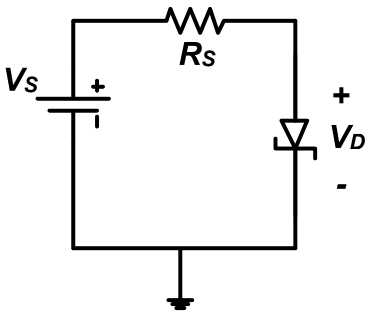


Fig 1

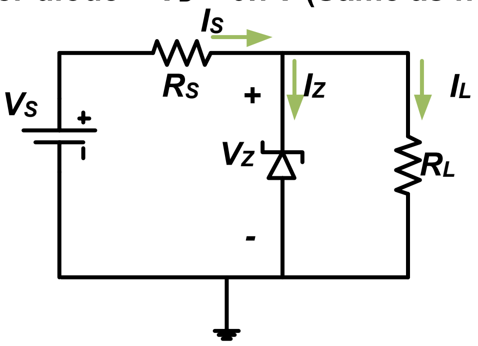
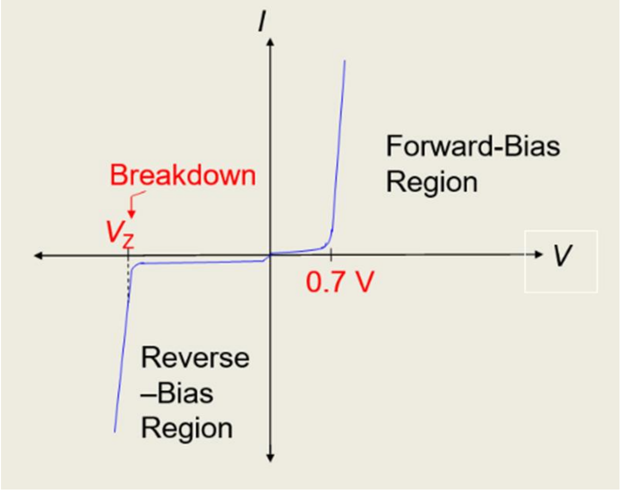
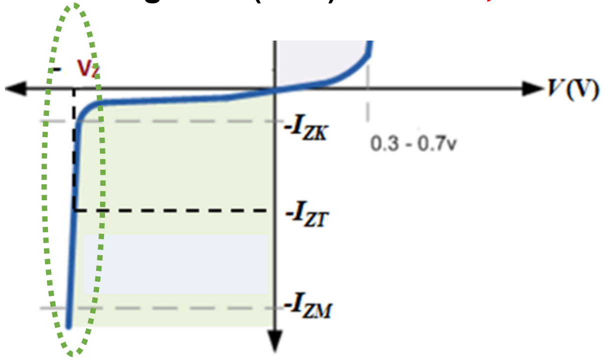


Fig 2



Reverse Bias (Fig 2) → If $V_S > V_Z$ and $I_{ZK} < I_Z < I_{ZM}$, then voltage across Zener Diode = V_Z

From Fig 2--→ (KCL) $I_S = I_Z + I_L$, $I_L = V_Z/R_L$, $^{\#}I_S = (V_S - V_Z)/R_S$

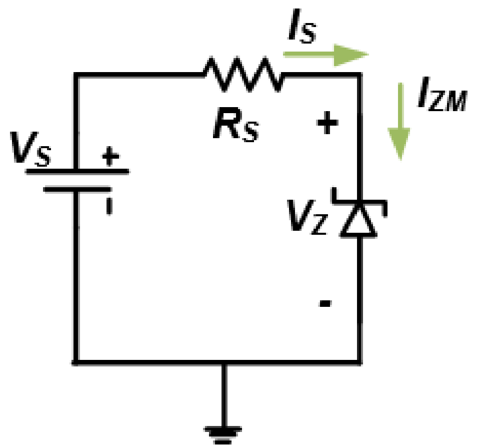


Zener Diode -
I-V Characteristic in Breakdown Region

- Zener diodes are specifically used in applications where the **reverse-breakdown region** is the desired operating region.
- If $I_Z < I_{ZK}$, Zener is **reverse biased** but **not breakdown**, Hence V_Z cannot be stabilized at a constant value.
- If $I_Z > I_{ZM}$, Zener diode **may be damaged** due to excessive power!

Selection of current limiting resistor, RS

To find minimum R_S to keep $I_Z < I_{ZM}$ → The condition $R_L = \infty$ (open, No Load) & $I_Z = I_{ZM}$

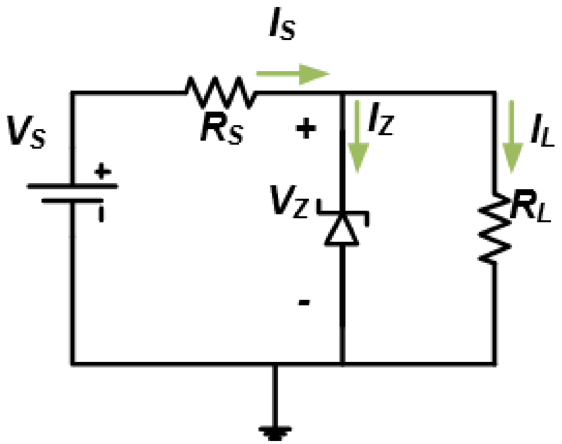


$$I_S = I_Z = I_{ZM}$$

$$I_S = \frac{V_S - V_Z}{R_{S(min)}}$$

$$R_{S(min)} = \frac{V_S - V_Z}{I_S}$$

If R_S and R_L values are given,



$$KCL \rightarrow \quad I_S = I_Z + I_L$$

$$I_S = \frac{V_S - V_Z}{R_S}$$

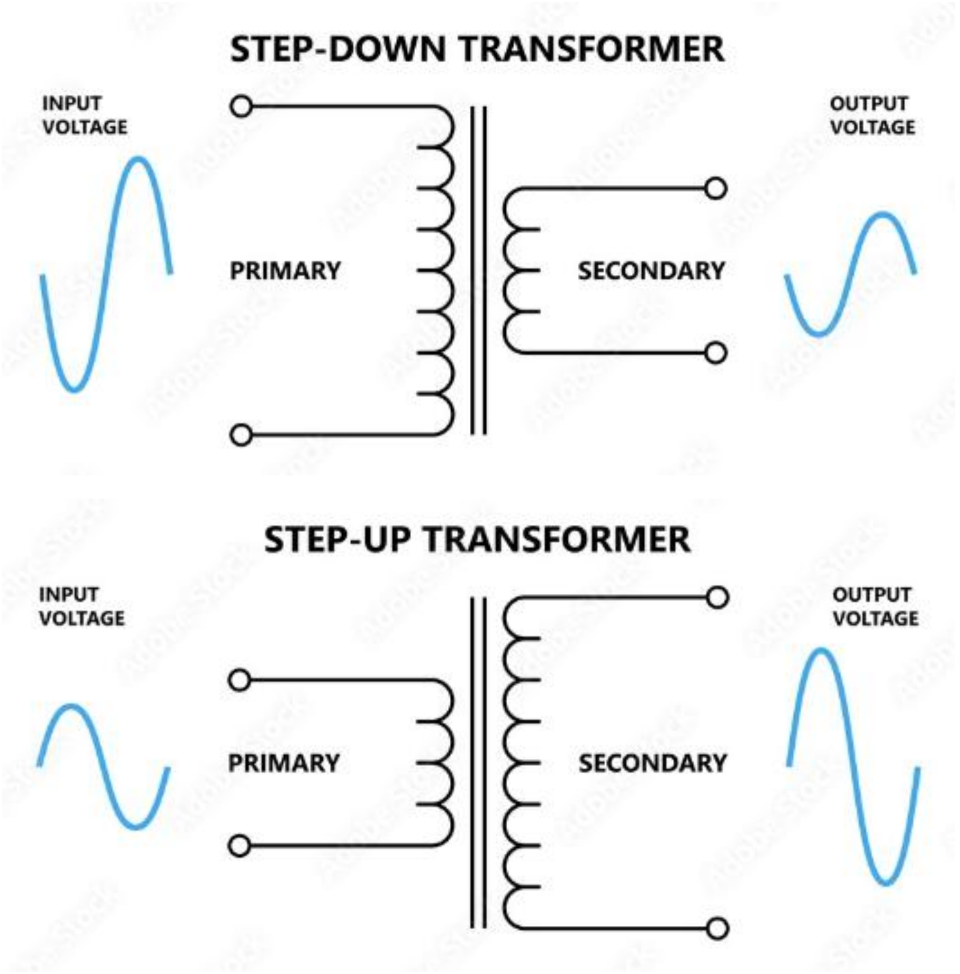
$$I_L = \frac{V_Z}{R_L}$$

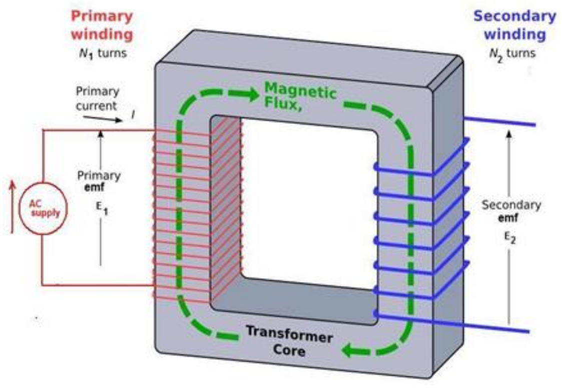
$$I_Z = I_S - I_L$$

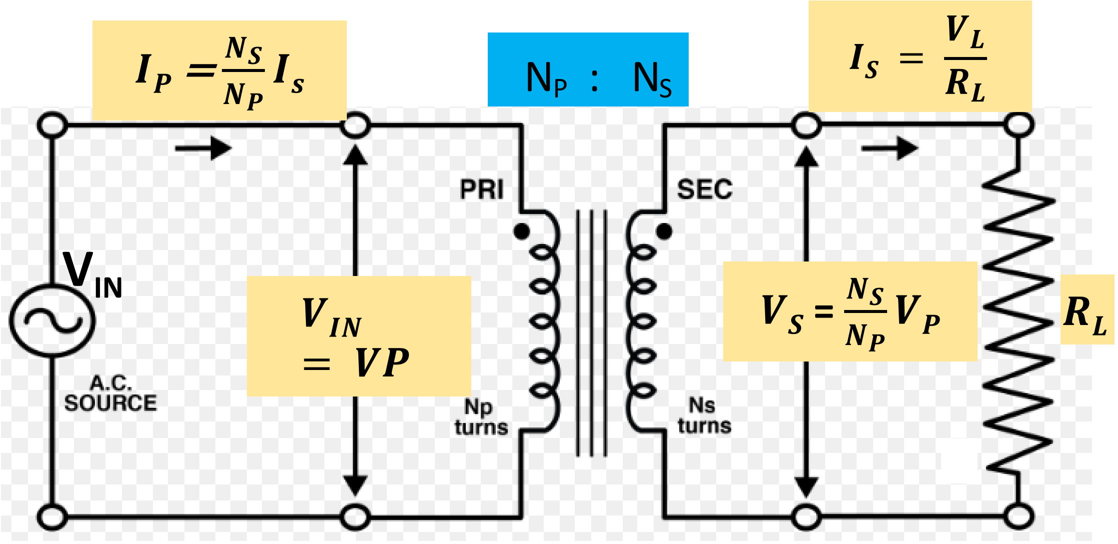
If $I_{ZK} < I_Z < I_{ZM}$, Zener Diode is functioning as Voltage Regulator.

- If $I_Z < I_{ZK}$, V_Z cannot be stabilized at a constant value. Zener Diode will not function as Voltage regulator.
- If $I_Z > I_{ZM}$, Zener diode **may be damaged** due to excessive power!

DC Power Supply: Transformer



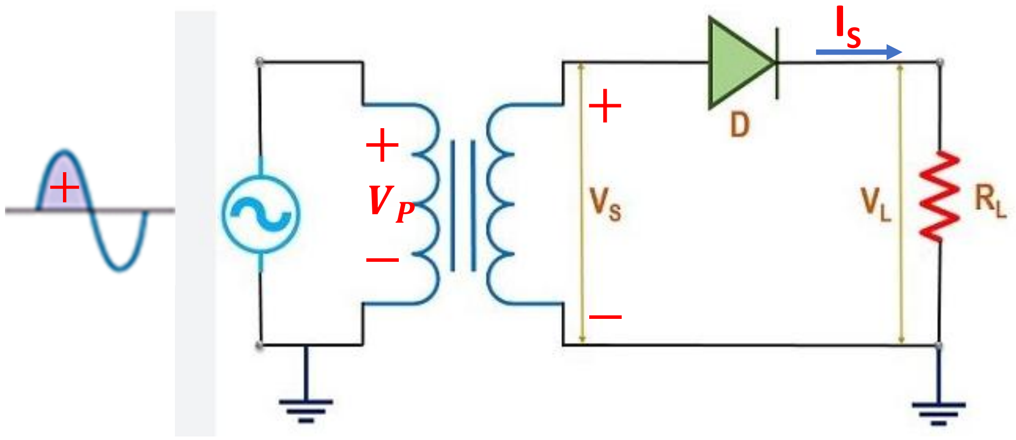


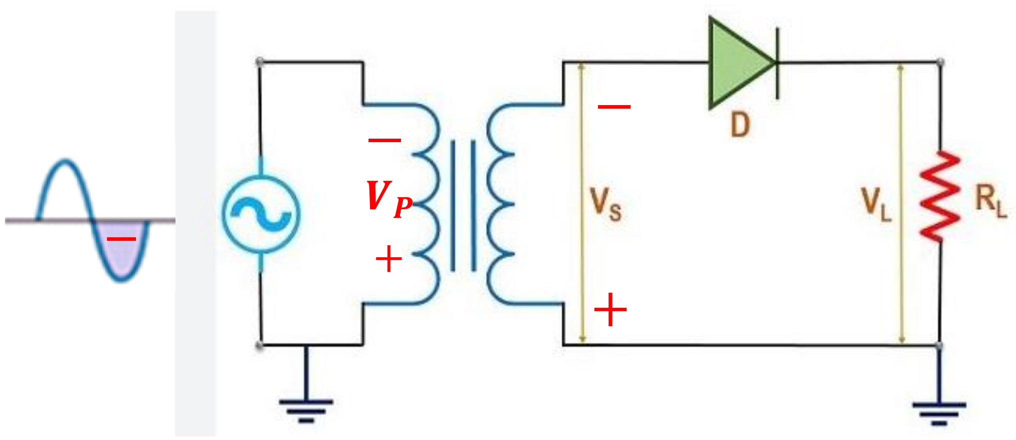


Schematic Symbol

$$(\text{turn ration})\ n = \frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P}$$

Half Wave Rectifier





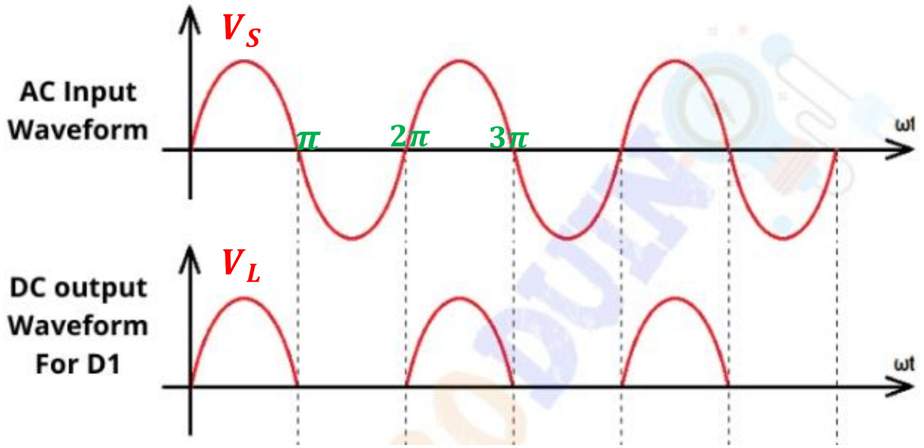
During the positive half-cycle, D_1 is **forward** biased.

KVL → $V_S = V_D + V_L$, where V_D =0.7V.

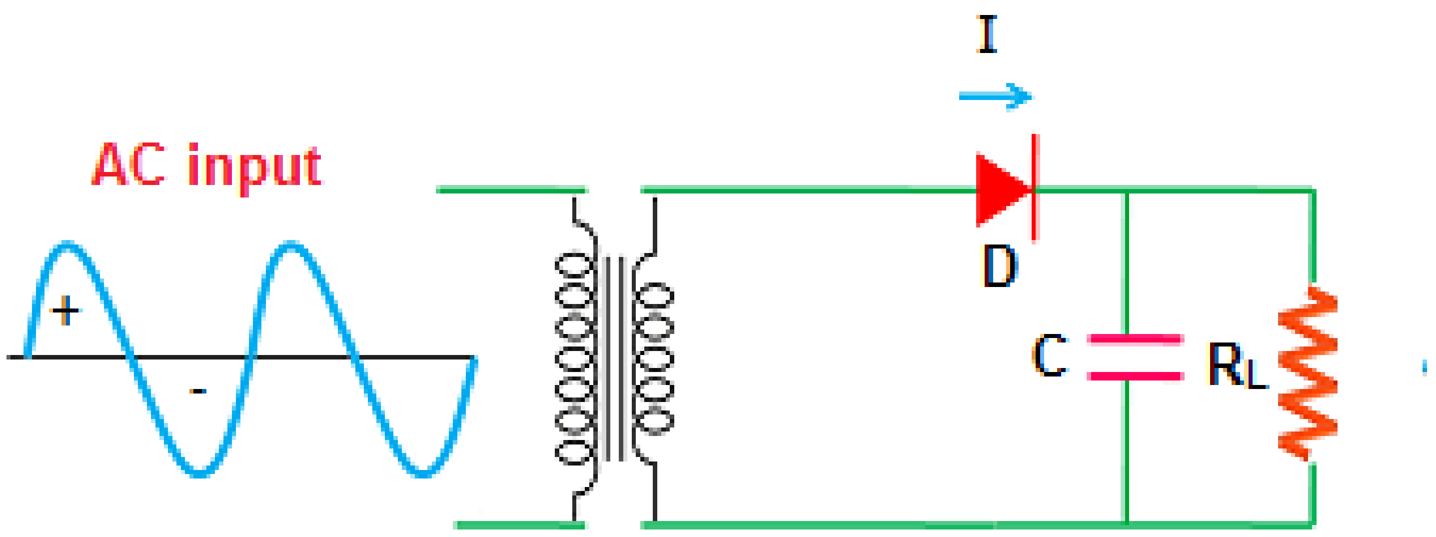
$$I_S = \frac{V_S - V_D}{R_L}, V_L = I_L \times R_L$$

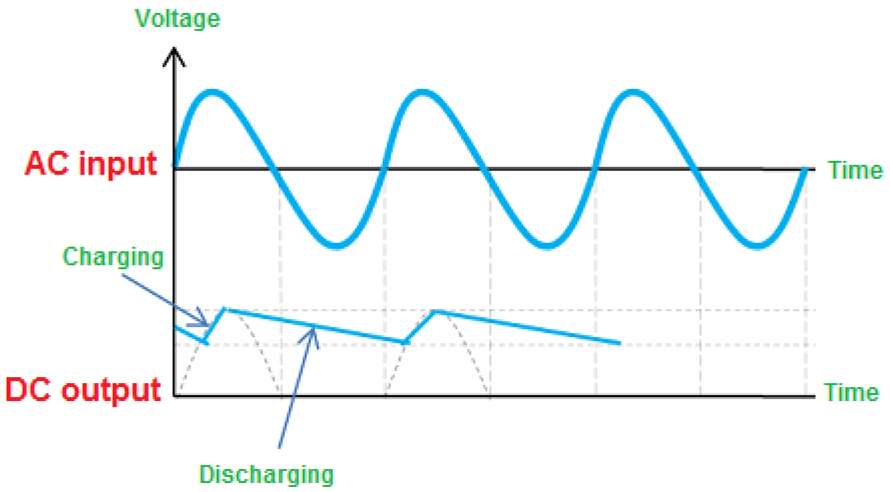
During the negative half-cycle, D_1 is **reverse** biased.

$$I_S = 0, V_L = 0$$

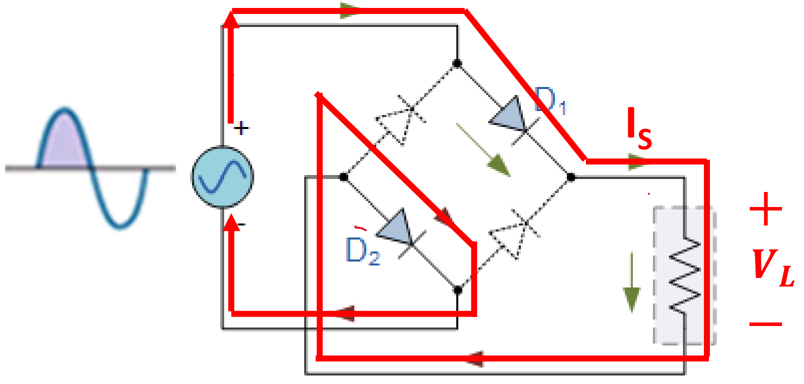


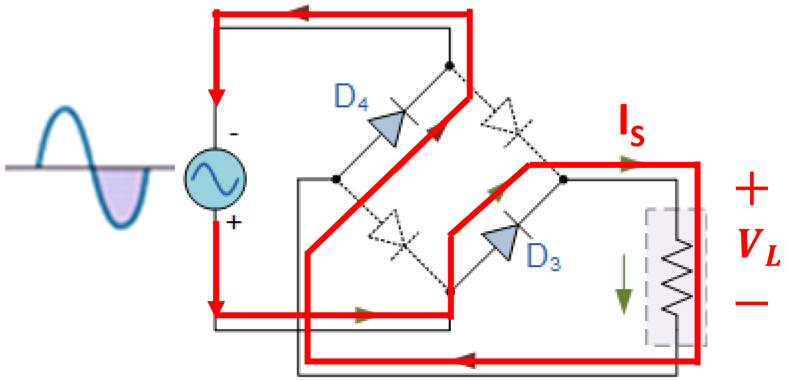
Half Wave Rectifier with Capacitor Filter





Full Wave Rectifier





During the positive half-cycle, D_1 & D_2 are **forward** biased and D_3 & D_4 are **reverse** biased.

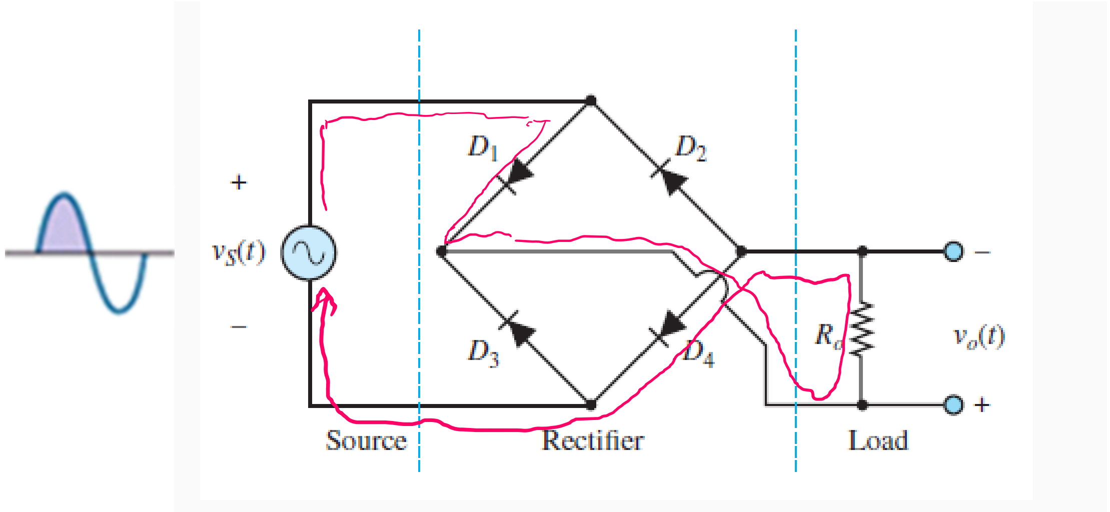
KVL → $V_S = V_{D1} + V_{D2} + V_L$,

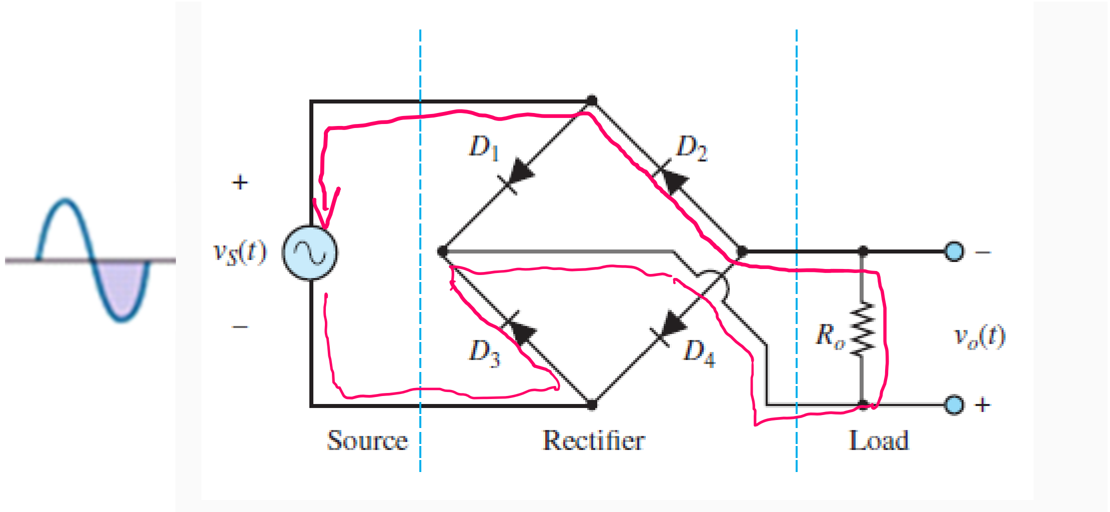
$$I_S = \frac{V_S - 2V_D}{R_L}, V_L = I_L \times R_L$$

During the positive half-cycle, D_1 & D_2 are **forward** biased and D_3 & D_4 are **reverse** biased.

KVL → $V_S = V_{D3} + V_{D4} + V_L$,

$$I_S = \frac{V_S - 2V_D}{R_L}, V_L = I_L \times R_L$$





During the positive half-cycle, D_1 & D_4 are **forward** biased and D_3 & D_4 are **reverse** biased.

KVL → $V_S = V_{D1} + V_{D4} + V_o$,

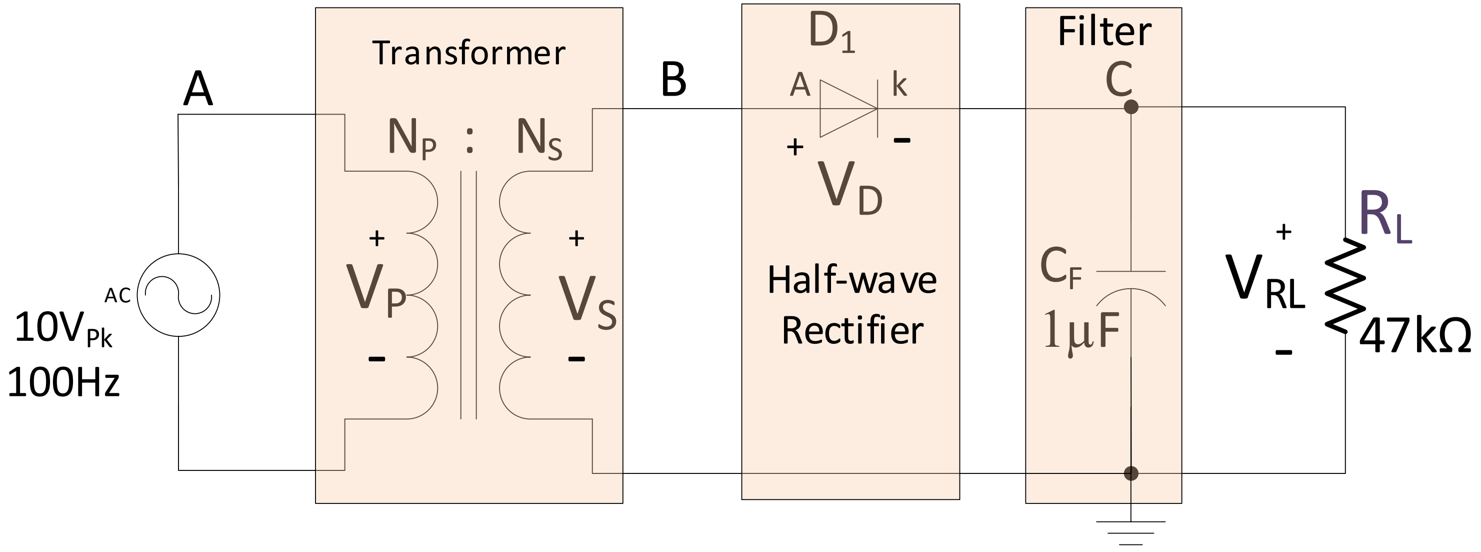
$$I_S = \frac{V_S - 2V_D}{R_O}, V_o = I_L \times R_O$$

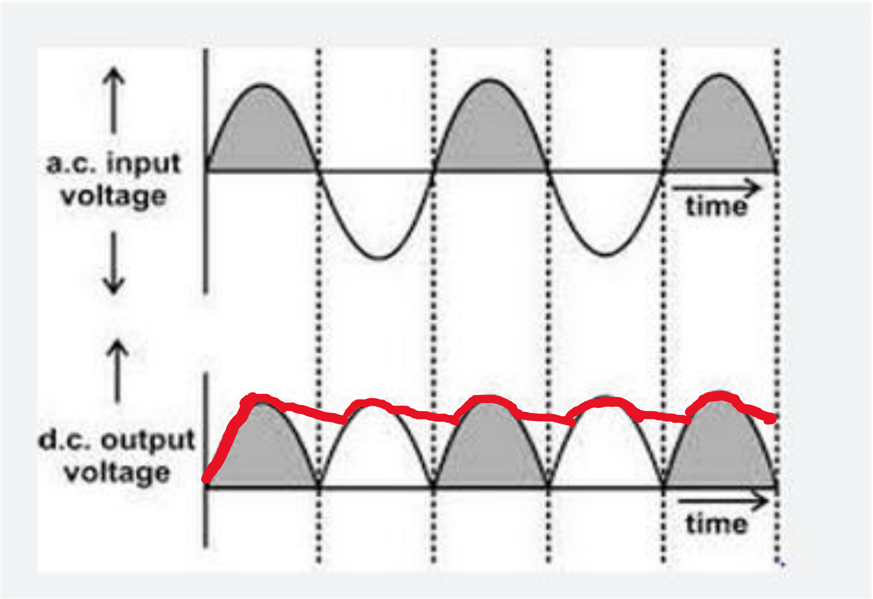
During the positive half-cycle, D_2 & D_4 are **forward** biased and D_3 & D_2 are **reverse** biased.

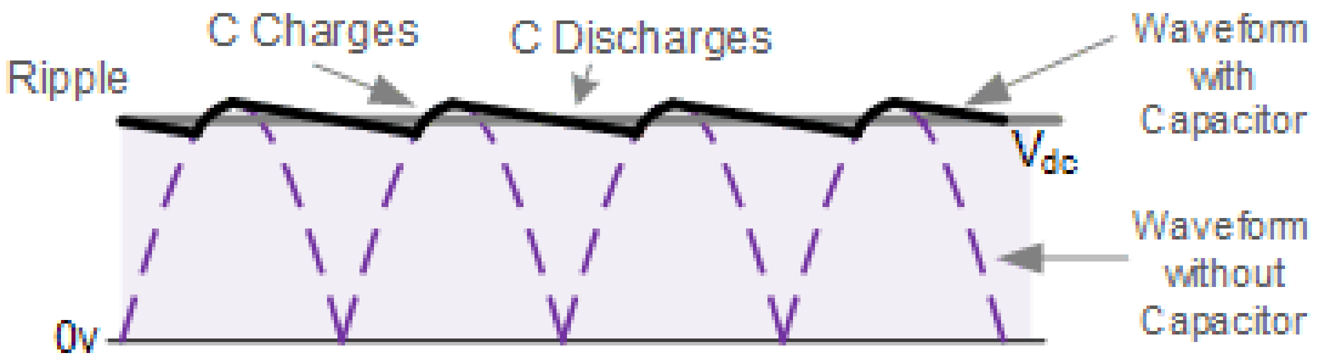
KVL → $V_S = V_{D3} + V_{D2} + V_L$,

$$I_S = \frac{V_S - 2V_D}{R_L}, V_o = I_L \times R_L$$

Effects of Filter Capacitor and Resistive load







Voltage Regulator